UNIT - II

DATA LINK LAYER DESIGN ISSUES:

- Datalink Layer is the second layer in OSI Reference model.
- DLL Supervises the flow of information between adjacent nodes.
- The data link layer uses the services of the physical layer to **send and receive bits over communication channels.**
- It has a number of functions, including:

1. Providing a well-defined service interface to the network layer.
2. Dealing with transmission errors.
3. Regulating the flow of data so that slow receivers are not swamped by fast senders.

The **Data Link Layer** is responsible for reliable communication between two directly connected devices. Its main **design issues** include:

Design Issues of the Data Link Layer

1. **Framing**

- Converts the continuous bit stream into frames.
- Adds headers and trailers to identify the beginning and end of each frame.
- Makes error detection and retransmission easier.

2. Physical Addressing

- Adds source and destination MAC addresses.
- Ensures frames reach the correct device within a local area network (LAN).

3. Error Control

- Detects errors caused during transmission.
- Uses CRC, checksums, parity bits, and acknowledgments.
- Supports retransmission of damaged or lost frames.

4. Flow Control

- Prevents a fast sender from flooding a slow receiver.
- Ensures efficient use of network resources.
- Methods include Stop-and-Wait and Sliding Window protocols.

5. Access Control (Medium Access Control)

- Controls access to a shared communication medium.
- Prevents collisions when multiple devices attempt to transmit simultaneously.
- Examples: CSMA/CD, CSMA/CA, Token Passing.

6. Reliable Delivery

- Guarantees that frames reach the destination correctly and in order.
- Uses sequence numbers, acknowledgments, and retransmissions.

All these are important functions for reliable communication and plays an important role in designing data link layer.

❑ Services provided to the Network layer:

- The primary function of the data link layer is to **provide services to the network layer.**
- The principal service is **transferring data from the network layer on the source machine to the network layer on the destination machine.**
- The **two Datalink layer communicates with each other by Datalink control protocols.**
- Following are the important services provided by the Data link layer to the Network layer.

1. **Unacknowledged connectionless service.**
2. **Acknowledged connectionless service.**
3. **Acknowledged connection-oriented service.**

1. Unacknowledged connectionless service:

- It is un acknowledgement service of transmission.
- Here the source machine **sends the data to the destination machine without any acknowledgement**.
- For this **no connection is either established or released.**
- If data is loss due to noise or interrupts the lost **data is not even recovered by the layer.**

2. Acknowledged connectionless service:

- It is acknowledgement connection less service of transmission.
- Here **each data frame is acknowledged by the destination machine.**
- If any **data frame is lost or not arrived in time the same message can be *transmitted again.***
- In this service no connections are used.

3. Acknowledged connection oriented service:

- It is acknowledgement connection oriented service of transmission.
- Here the **connection is established for data transmission.**
- Each frame is **numbered before transmission and corresponding acknowledgement is also received.**
- Here connection is established before transmission of data.

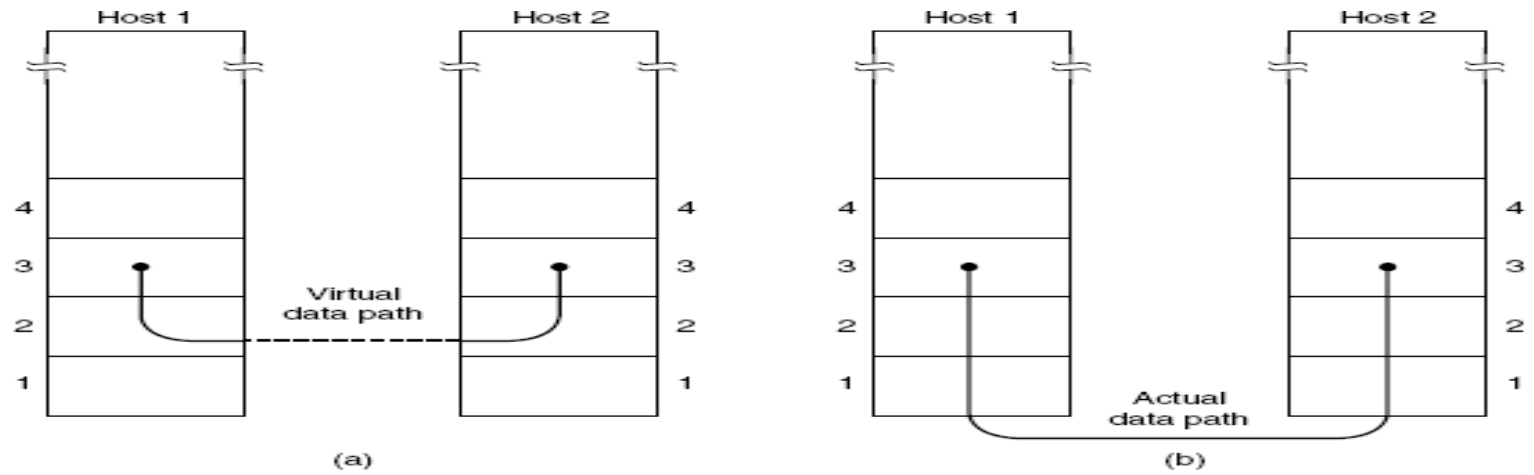


Figure 3-2. (a) Virtual communication. (b) Actual communication.

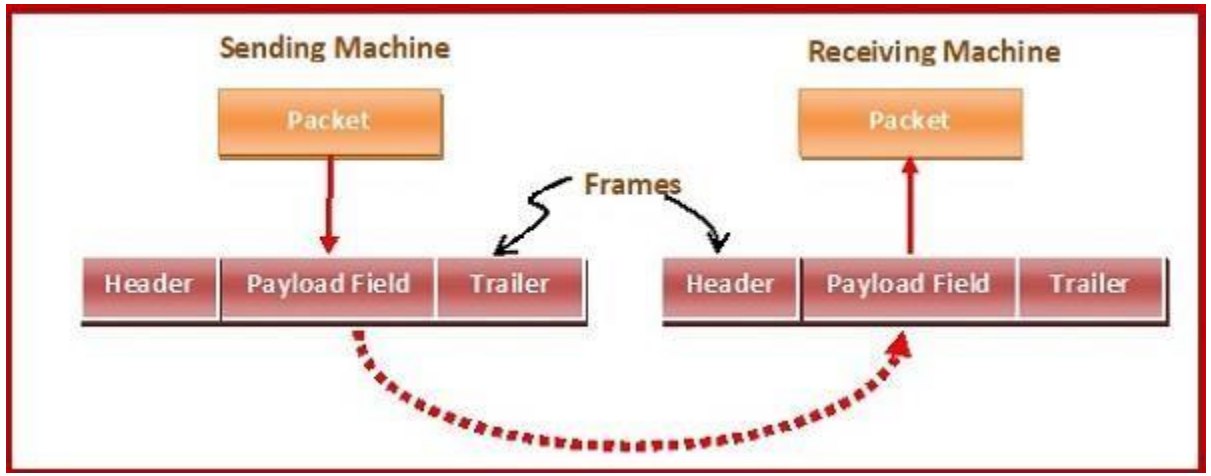
❑ Framing:

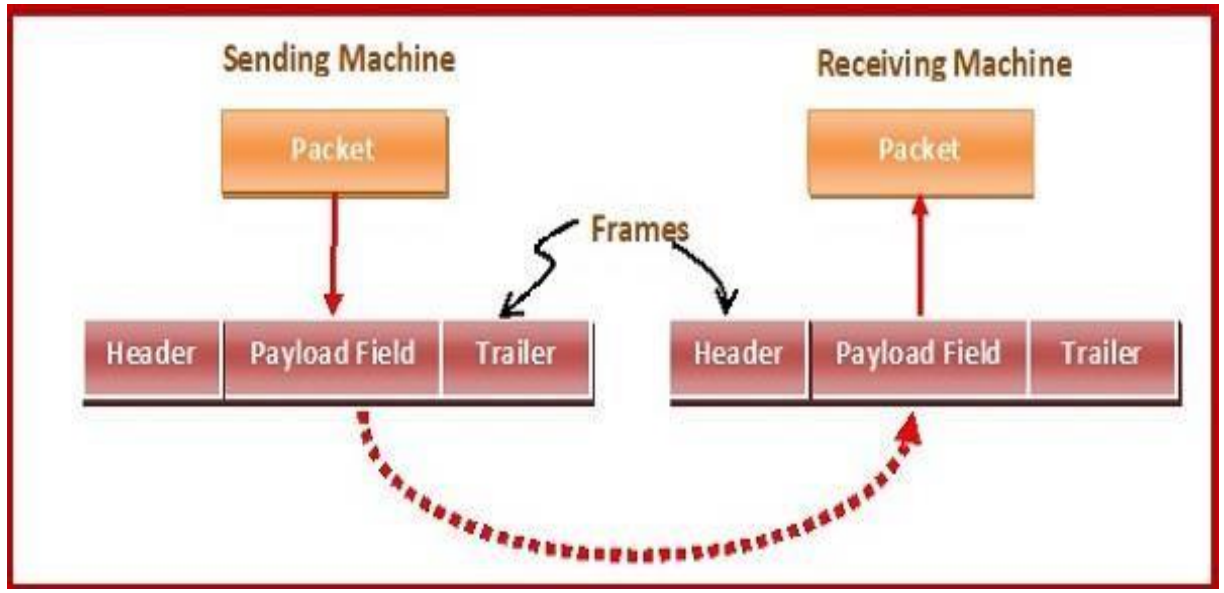
- ***"Breaking of the bit stream by inserting spaces or time gaps is called framing".***

- To service the **network layer , data link layer uses the service provided to the physical layer.**
- The data link layer encapsulates each data packet from the network layer into frames that are then transmitted.

A frame has three parts, namely −

•Frame Header
•Payload field that contains the data packet from network layer
•Trailer





•**Frame Header**: It contains the source and the destination addresses of the frame and the control bytes.

•**Payload field**: It contains the message to be delivered.

•**Trailer**: It contains the error detection and error correction bits (damaged or not). It is also called a Frame Check Sequence (FCS).

•**Flag**: Two flag at the two ends mark the beginning and the end of the frame.

- **Physical layer accepts the bit stream and deliver to the destination.**

- **This bit stream (sequence of bits) may contain errors i.e. number of bits received may not be equal to the number of bits transmitted.**

- **The data link layer breaks the stream into discrete frames (individual frames) and compute the check sum for each frame.**

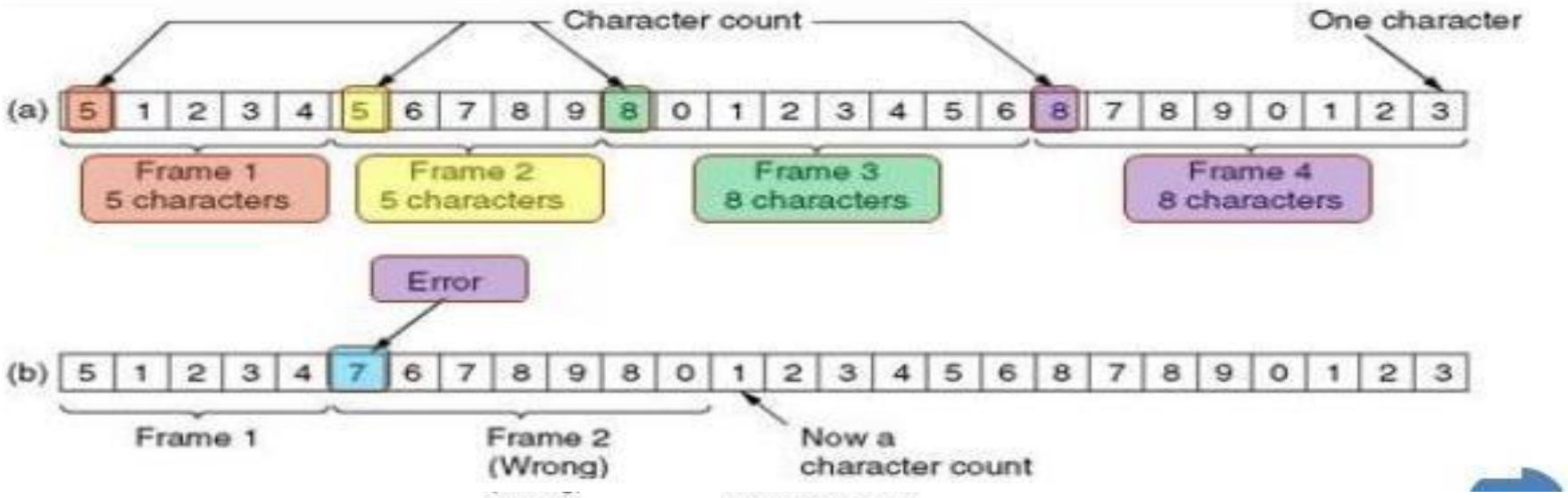
- *So it is difficult and risky to count each frame , so by using the following methods to overcome*

    1. ***Character count***
    2. ***Starting and Ending characters with character stuffing.***
    3. ***Starting and Ending flags with bit stuffing.***

1. Character count:

The first framing method uses a field in the **header** to **specify the number of characters in the frame**. When the data link layer at the destination sees the character count, it knows how many characters follow and hence where the end of the frame is.

This technique is shown in Fig for four frames of sizes 5, 5, 8, and 8 characters, respectively.
Figure. A character stream. (a) Without errors. (b) With one error.



Dis advantage For example, if the character count of 5 in the second frame becomes a 7.

2. Starting and Ending characters with character stuffing:

- *Each frame starts and ends with a special byte (i.e. FLAG byte).*
- *If already FLAG byte occurs in the frame Stuff (added) an Extra byte (i.e. ESC).*
- *If Esc byte is there in the frame add one more Esc in the frame.*
- *If it consist of both Flag and Esc add one more Esc and Flag bytes in the given frame.*

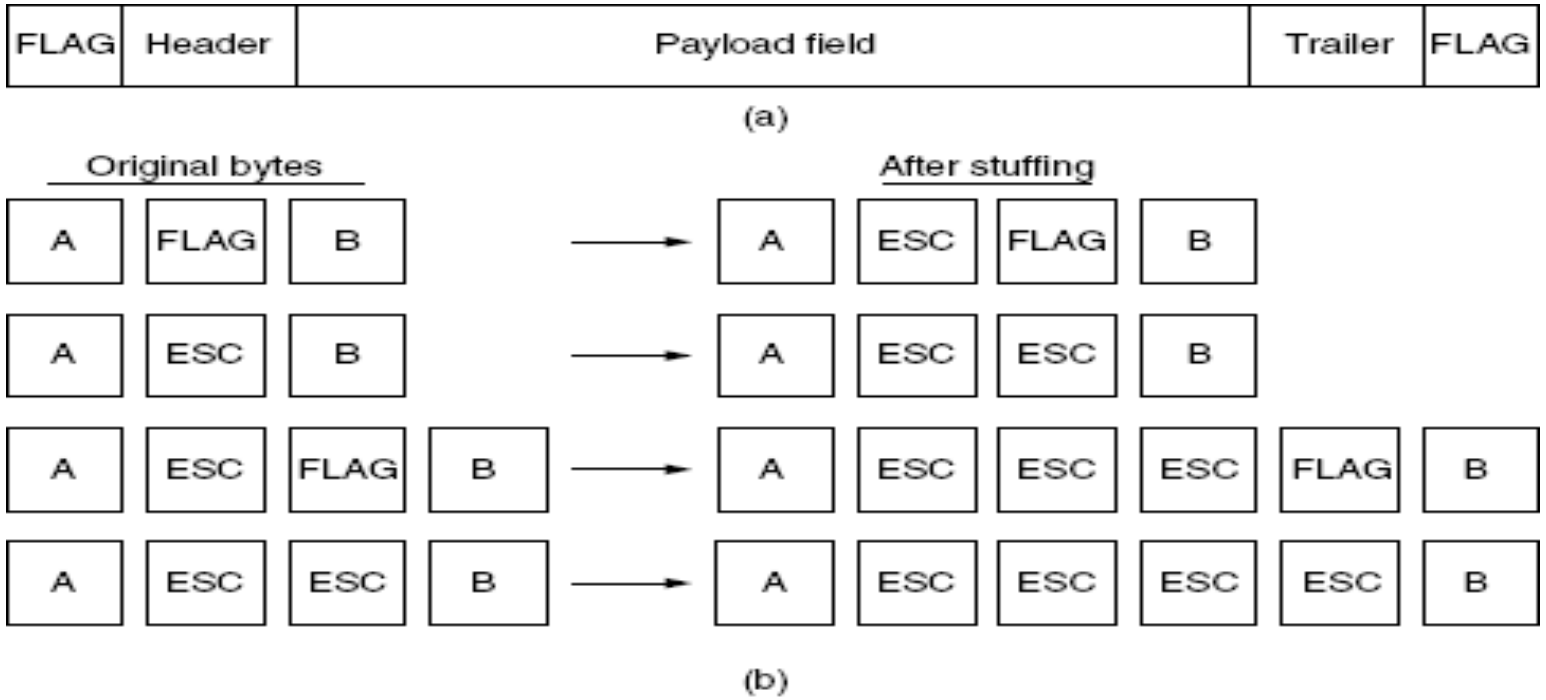
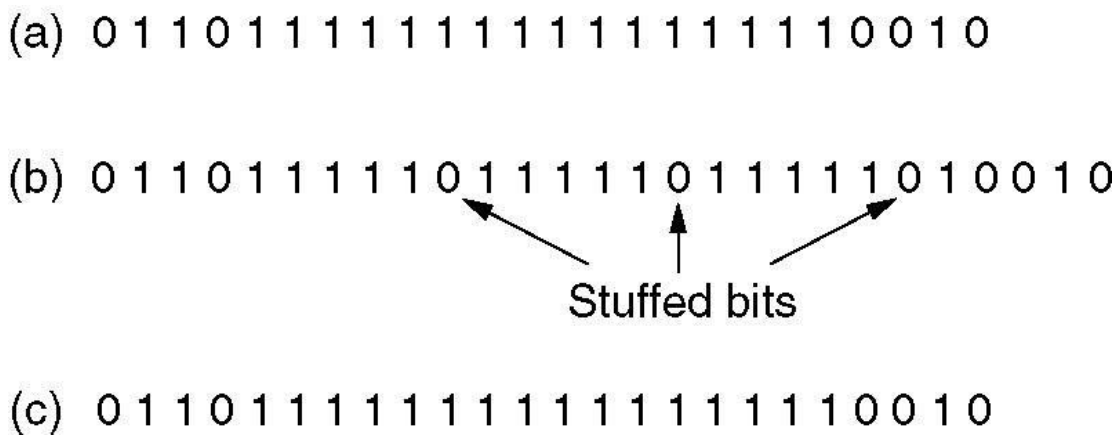


Figure 3-4. (a) A frame delimited by flag bytes. (b) Four examples of byte sequences before and after byte stuffing.

3. Bit Stuffing:

In bit stuffing a specific bit is stuffed into the out going character stream.
In this example if any 5 consecutive 1 s are encountered in the data it is automatically stuffs a 0 bit is replaced in the frame.

Each frame begins &Ends a Special bit pattern 01111110 when even sender sees 5 consecutive 1 s it stuffs 0.



Bit stuffing

(a) The original data.

(b) The data as they appear on the line.

(c) The data as they are stored in receiver's memory after de stuffing.

Error Control:

- Only technique in error control is **Retransmission.**
- **To ensure proper sequencing and safe delivery of frames at the destination , An Acknowledgement should be sent by the destination network.**
- **The receiver sends back positive or negative acknowledgements about incoming frames.**
- **If the sender receives a positive Acknowledgement it means the frame has arrived safely.**
- **If the sender receives a Negative Acknowledgement it means the some thing wrong in that frame.**

- **Phases of Error control**

  - **Error Detection (find the errors).**
  - **Acknowledgement (ACK).**
    - **Positive ACK**
    - **Negative ACK.**
    - **Retransmission.**

Flow Control:

- When **sender is running on Fast machine and Receiver is on slow receiver (overloaded data) some data may loss.**

- To solved the problem of marking the start and end of each frame how to make sure all frames are eventually **delivered to the network layer at the destination and in the proper order.**

- **Both sender and receiver maintain the same speed i.e. same capabilities.**

- This situation can occur when the sender is running on a fast, powerful computer and the receiver is running on a slow, low-end machine.

- Clearly to prevent this situation **Two approaches are commonly used.**

1). In the first one, **feedback-based flow control**, the receiver sends back information to the ***sender giving it permission to send more data***, or at **less data** telling the sender how the receiver is doing.

2). In the second one, **rate-based flow control**, the protocol has a built-in mechanism that **limits the data transfer rate at which senders may transmit data.**

Error Correction and Error Detection:

- During transmission of data from the source to a receiver many factors like noise, change some part of data , many types of interrupts in the data.

- So **system to detect and correct such errors.by using these two techniques.**

- **“Error detection”** techniques is ability to **detecting** such **errors due to** causes noise or interrupts etc..

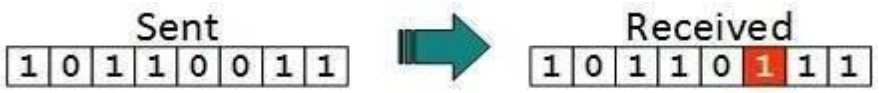
- **“Error Correction”** techniques is enables reconstruction of the original data in many cases.

- **Redundancy :** Error detection uses the concept of Redundancy . Redundancy means adding extra bits for Detecting errors.

Types of Errors:

There may be three types of errors:

- **Single bit error**



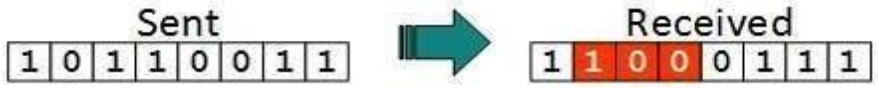
In a frame, there is only one bit, anywhere though, which is corrupt.

- **Multiple bits error**



Frame is received with more than one bits in corrupted state.

- **Burst error**



Frame contains more than one consecutive bits corrupted.

Basic approach used for error detection is the use of redundancy bits, where **additional bits are added to facilitate detection of errors.**

Error Detection techniques are:

1. Parity check (1. Simple parity , 2. Two - Dimensional parity).
2. Checksum.
3. Cyclic redundancy check (CRC).

1.Parity :

Simplest form of error detection is to append a single bit called Parity bit to a string of data. There are 2 forms of parity check methods.

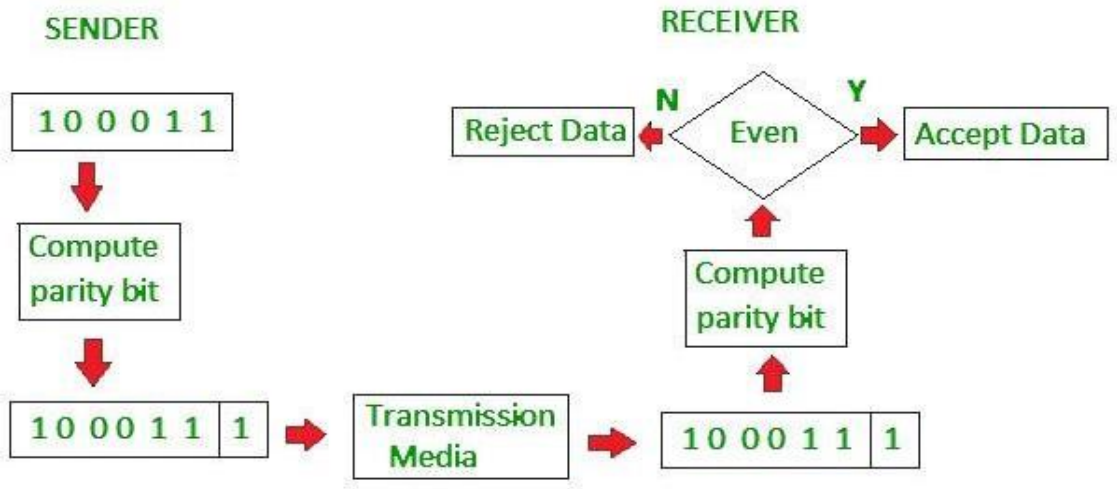
1. **Simple parity**
2. **Two - Dimensional parity**

1. Simple parity:

Blocks of data from the source are subjected to a check bit or parity bit generator form, where a parity of :

- 1 is added to the block if it contains odd number of 1's, (Odd Parity)
- 0 is added if it contains even number of 1's. (Even parity)

This scheme makes the total number of 1's even, that is why it is called even parity checking.



Detection -examples

Example 1

Suppose the sender wants to send the word *world*. In ASCII the five characters are coded as

1110111 1101111 1110010 1101100 1100100

The following shows the actual bits sent

1110111**0** 1101111**0** 1110010**0** 1101100**0** 1100100**1**

Example 2

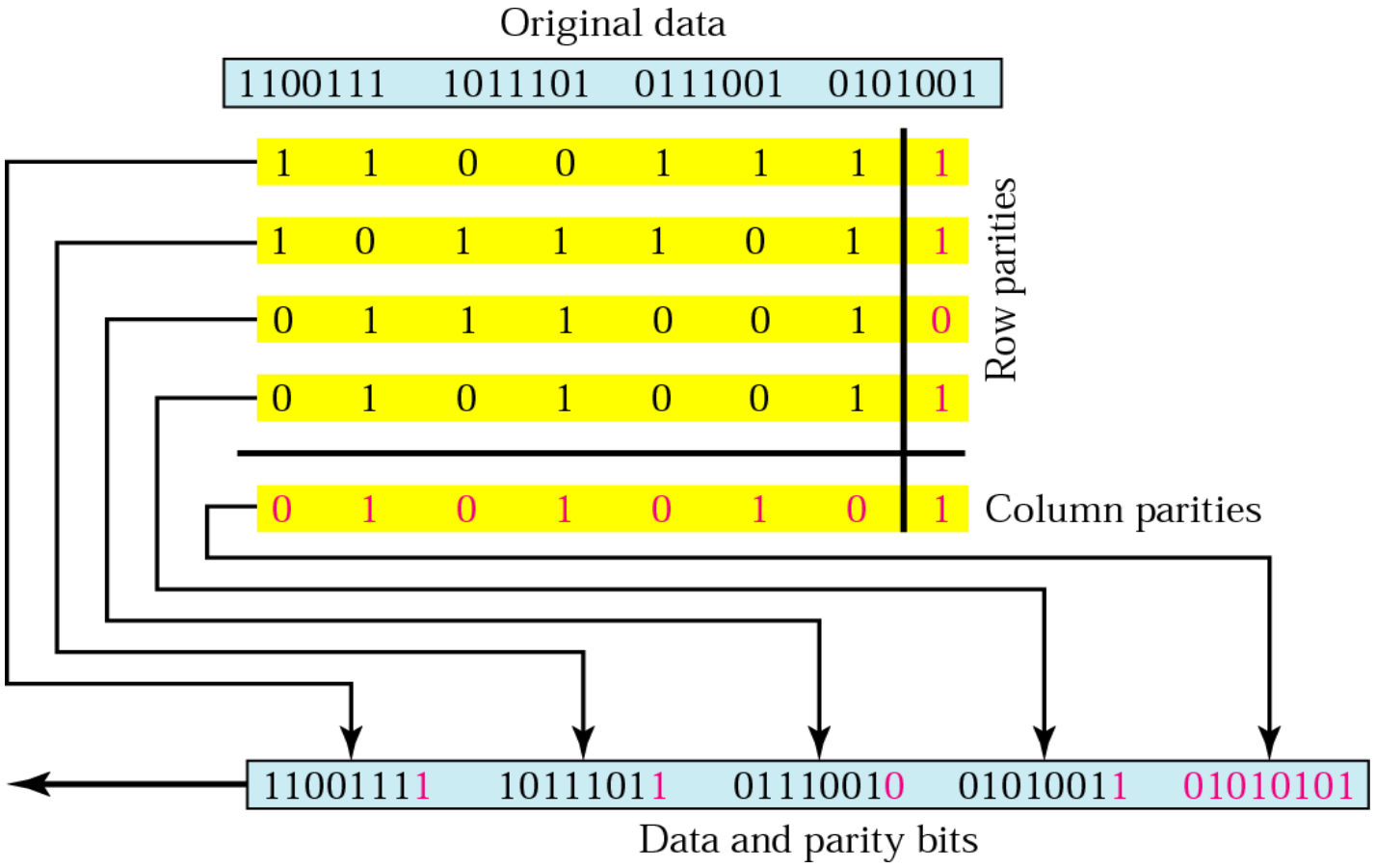
Now suppose the word is corrupted during transmission.

11111110 11011110 11101100 11011000 11001001

The receiver counts the 1s in each character and comes up with even and odd numbers (7, 6, 5, 4, 4). The receiver knows that the **data are corrupted, discards them, and asks for retransmission.**

2. Two –Dimensional Parity Check:

- In this 2 dimensional parity checking a block of bits is organized in **Rows and Columns (Table).**
- Then parity of each unit is calculated.
- Calculate parity bits for each row and column and attach to the original data and send to the receiver.



2. Check sum:

For **error detection by checksums, data is divided into fixed sized frames or segments**.

•**Sender's End** − The sender adds the segments using 1's complement arithmetic to get the sum. It then complements the sum to get the checksum and sends it along with the data frames.

•**Receiver's End** − The receiver adds the incoming segments along with the checksum using 1's complement arithmetic to get the sum and then complements it.

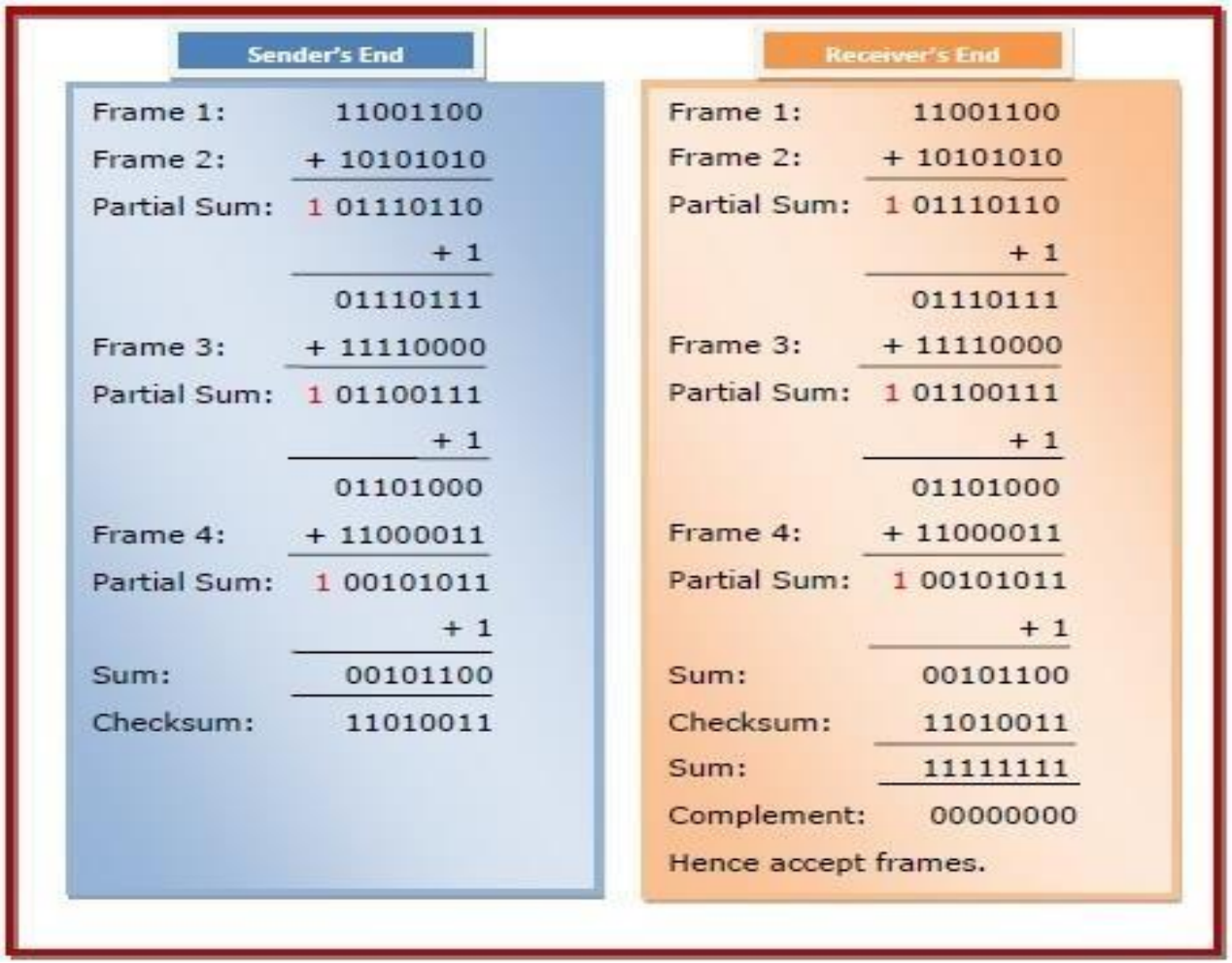
If the result is zero, the received frames are accepted; otherwise they are discarded.

Example

Suppose that the sender wants to send 4 frames each of 8 bits, where the frames are

11001100, 10101010, 11110000 and 11000011.

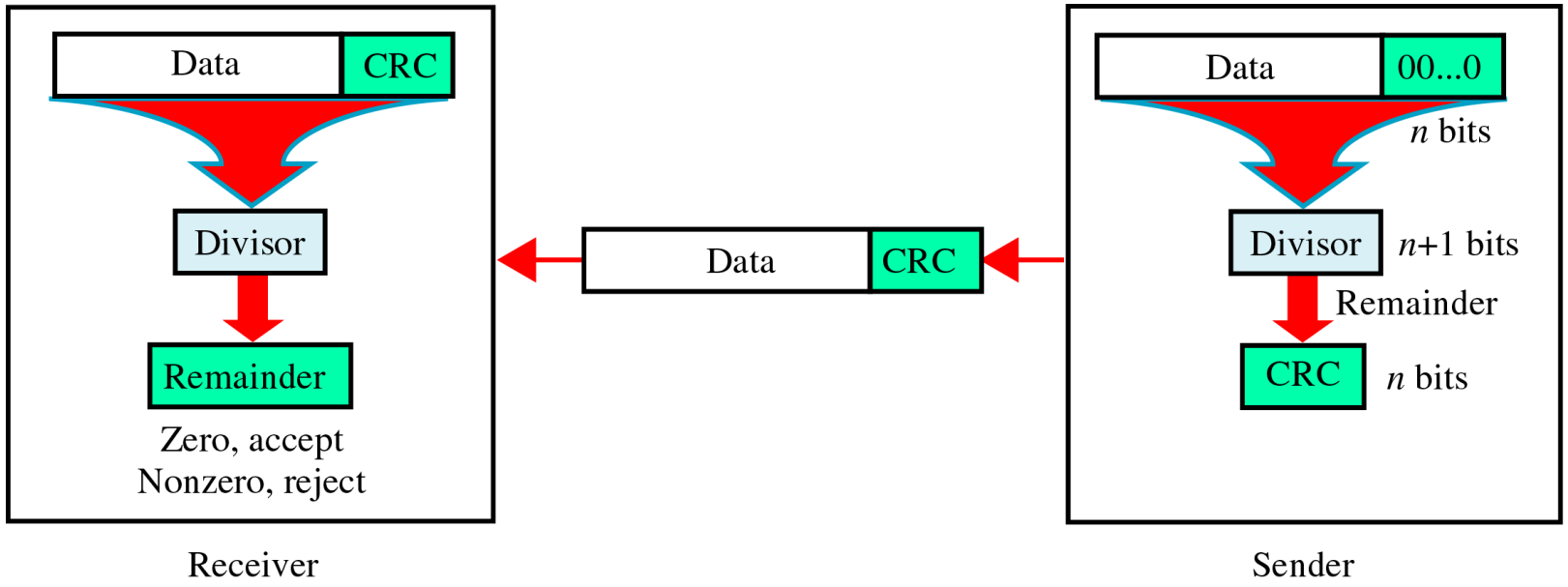
- The sender adds the bits using 1s complement arithmetic. While adding two numbers using 1s complement arithmetic, if there is a carry over, it is added to the sum.
- After adding all the 4 frames, the sender complements the sum to get the checksum, 11010011, and sends it along with the data frames.
- The receiver performs 1s complement arithmetic sum of all the frames including the checksum. The result is complemented and found to be 0. Hence, the receiver assumes that no error has occurred.



3. CRC (Cyclic Redundancy Check): Multiple bits verifies(brust errors)

- Data * CRC Generator (find the redundant bits) * CRC bits or redundant (added) bits.
- CRC Generator (**n** bits)
- CRC Bits (n-1 bits)
- CRC is based on binary division instead of addition bits as in parity check.
- In CRC a sequence of redundant bits (CRC Remainder) is appended to the data unit.
- So that it is exactly divisible by a second number .

- At the destination end the received data is divided by same number .
- If any remainder is generated if it is all 0's Success other wise Rejected . xor



- Data 100100
- CRC generator 1101
- CRC bits (n-1) 4-1=3 (000)initially
- Division XOR

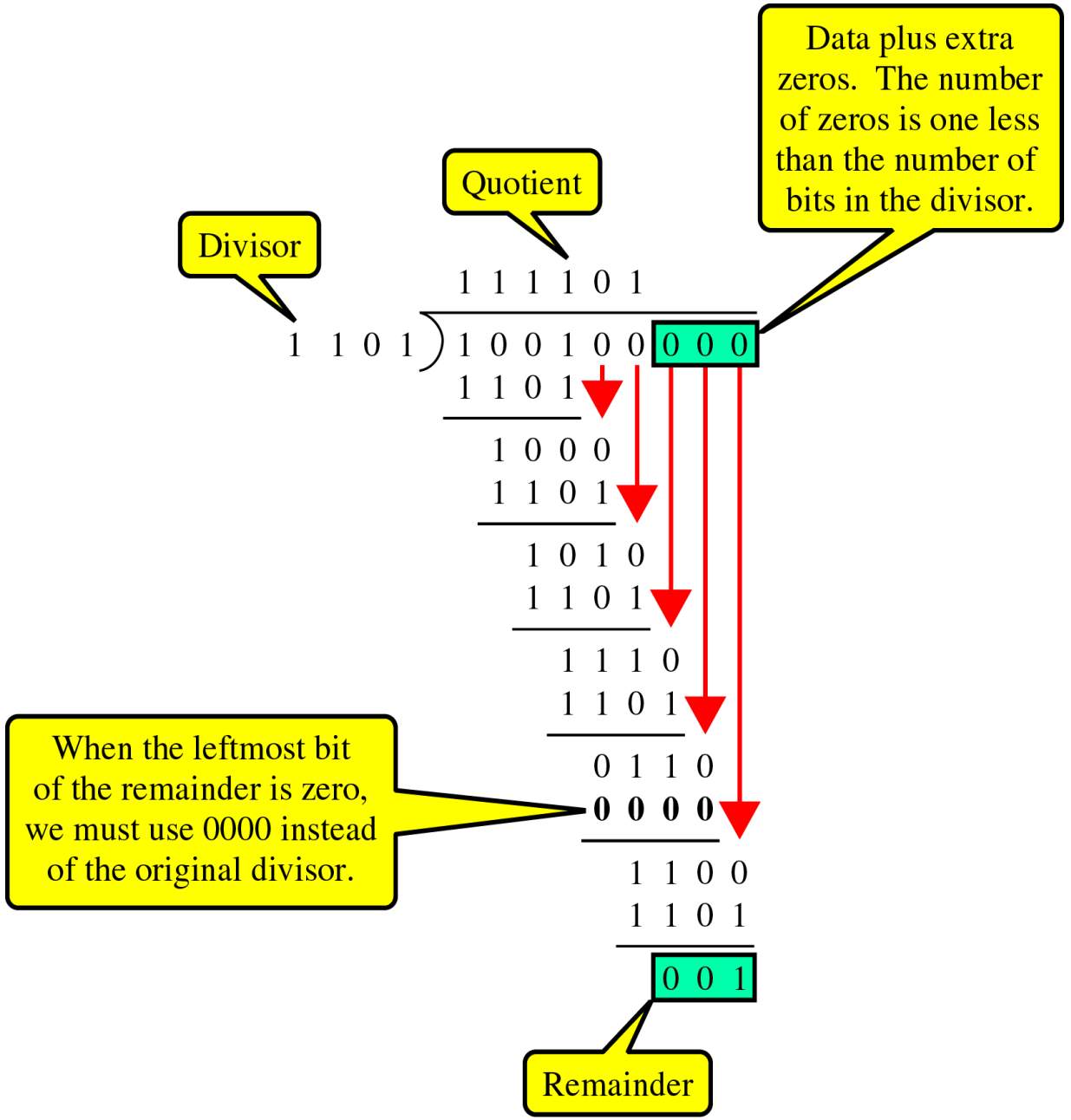
- Data : Polynomial expression

  10100111

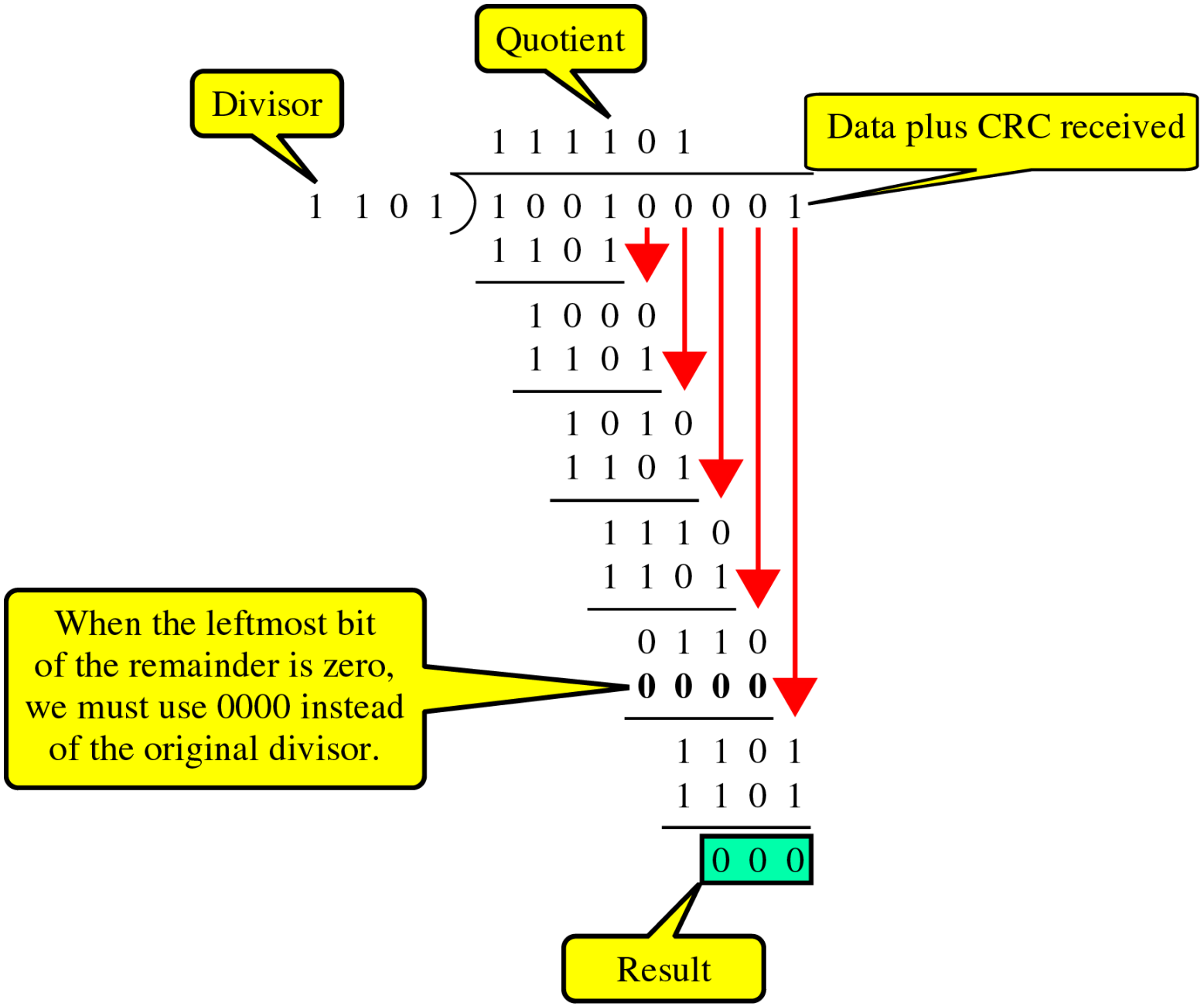
$$x^7 + x^5 + x^2 + x + 1$$

Transmitted Frame: 100100 001

- **Sender Data**



Receiver side verifies



Elementary Data link protocols:

Protocols in the data link layer are designed so that this layer can perform its basic functions: framing, error control and flow control.

"Framing" is the process of dividing bit - streams from physical layer into data frames .

"Error control" mechanisms deals with transmission errors and retransmission of corrupted and lost frames.

"Flow control" regulates speed of delivery and so that a fast sender does not drown a slow receiver.

Types of Data Link Protocols :

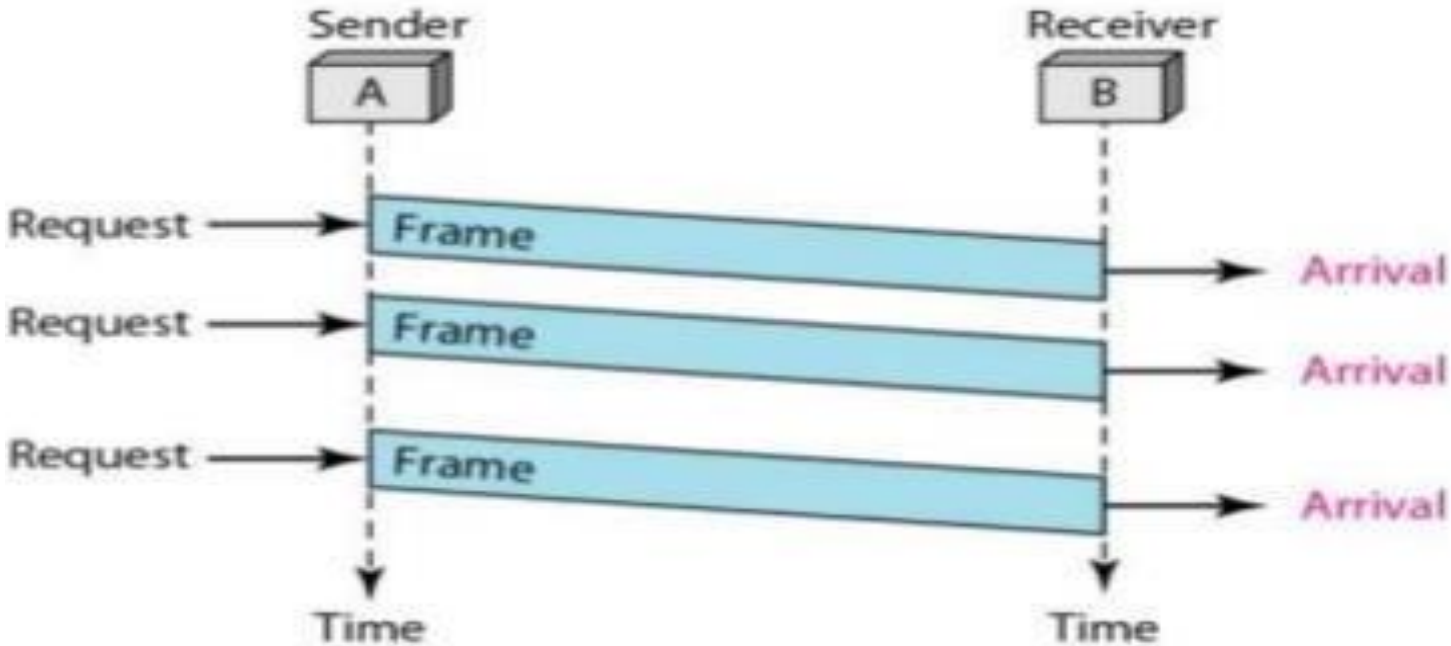
1. **Un Restricted protocol.**
2. **Simplex stop – and – wait protocol.**
3. **Simplex protocol for Noisy channel**

1. Un Restricted protocol :

- Data are transmitted in one direction only.
- Both transmitting and receiving are always ready state.
- Processing time can be ignored.
- Infinite buffer space available.
- The communication channel b/w sender and receiver never damage or never loss frames.
- No errors occur i.e. No frames damaged and No data loss.
- Advantages : simple protocol.
- Dis Advantage : If Continuous Data Transmits (**flooding** used for flow control) to the receiver congestions may occur data may loss.

Key characteristics include:

- Unidirectional data flow
- No flow control mechanisms
- Assumes error-free transmission
- Unlimited buffer capacity at receiver


Sender
A
Receiver
B
Request
Frame
Arrival
Request
Frame
Arrival
Request
Frame
Arrival
Time
Time

2. Simplex Stop and wait protocol :

- One directional flow of data from sender to receiver.
- Receiver has only finite buffer capacity.
- Communication is error free communication.(no data loss).
- Receiver having finite processing speed.
- Protocols in which the sender sends one frame and then waits for an acknowledgement before processing is called stop and wait protocol.

The **simplex stop and wait protocol** addresses the limitation of receiver processing speed. While data still flows in one direction, this protocol introduces flow control to prevent the sender from overwhelming the receiver.

The protocol operates through the following steps:

- **Step 1** – Sender transmits a data frame
- **Step 2** – Sender waits for acknowledgment (ACK) from receiver
- **Step 3** – Receiver processes the frame and sends ACK
- **Step 4** – Upon receiving ACK, sender transmits the next frame

- **Such elementary protocols are also called PAR (Positive Acknowledgment with Retransmission) or ARQ (Automatic Repeat reQuest).**
- **Data frames are transmitted in one direction (simplex protocols) where each frame is individually acknowledge by the receiver by a separate acknowledgment frame.**
- **The sender transmits one frame, starts a timer and waits for an acknowledgment frame from the receiver before sending further frames.**
- **A time-out period is used where frames not acknowledged by the receiver are retransmitted automatically by the sender.**
- **Frames received damaged by the receiver are not acknowledged and are retransmitted by the sender when the expected acknowledgment is not received and timed out.**

Advantage :

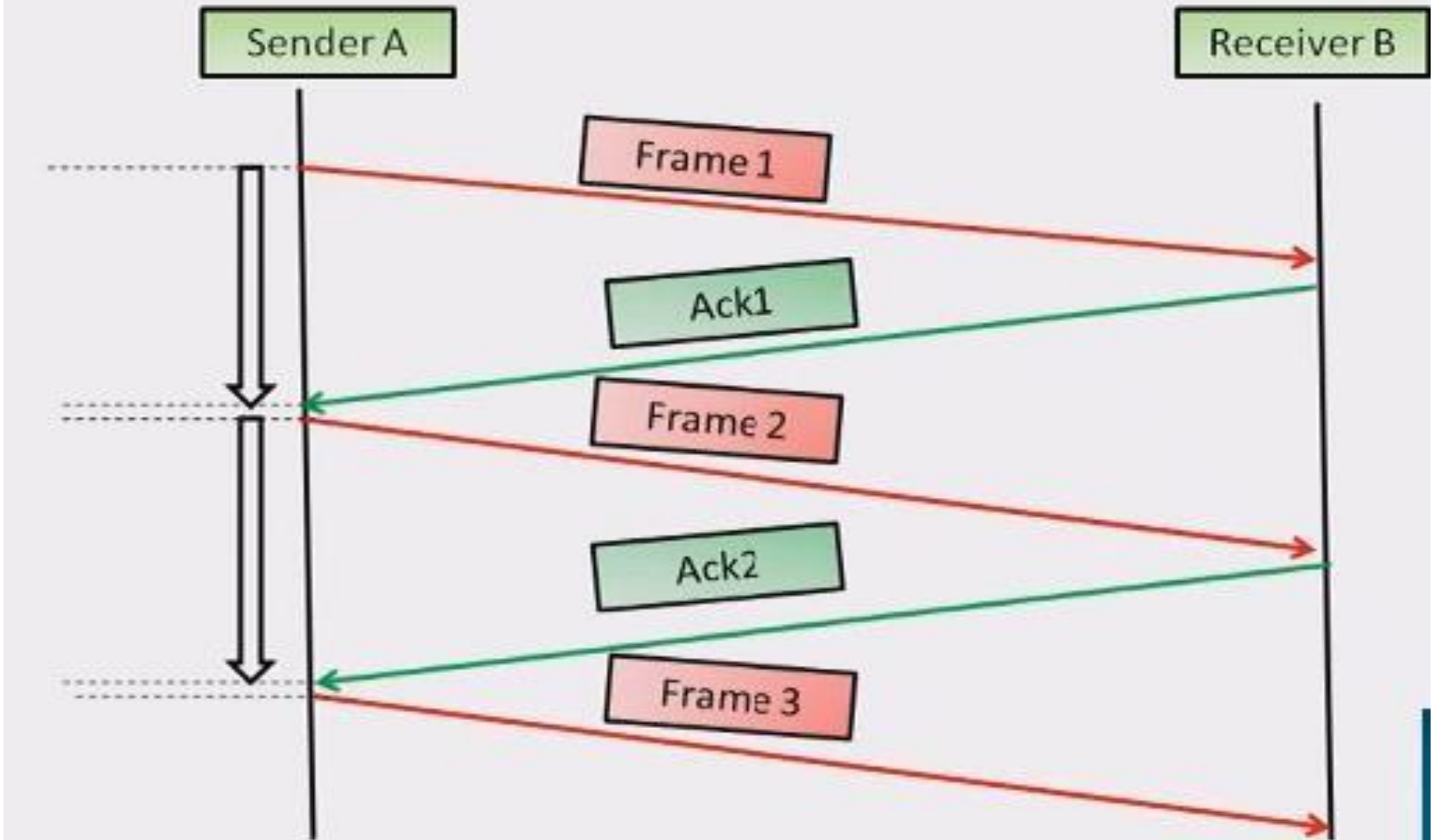
- Simple Reliable transmission.(no loss of data)**.**

Dis Advantage :

- Sender always wait for Acknowledgement frame to send next frame.(only one to send at a time it takes more time , if acknowledgement receives then only it transmits next frame.

- If Acknowledgement frame gets corrupted the sender cannot perform no more trans tractions.

Data Communication through Simplex Stop-and-Wait Protocol



3. Simplex protocol for Noisy channel:

Simplex Stop – and – Wait protocol for noisy channel is data link layer protocol for data communications with error control and flow control mechanisms. It is popularly known as Stop – and –Wait Automatic Repeat Request (Stop – and –Wait ARQ) protocol. It adds error control facilities to Stop – and – Wait protocol.

- Data transfer in only one direction.
- Finite processing capacity to the sender and receiver.(same processing speed).
- It is a Noisy channel , Errors in the data frame and acknowledgement frame are expects.
- Every frame has separate sequence number.
- After frame has to be transmitted Timer is started for finite time , Before the timer expires if the Acknowledgement is not received the frame again retransmits.

The **simplex protocol for noisy channels** is the most sophisticated of the three elementary protocols. It handles real-world conditions where data frames and acknowledgments can be corrupted or lost during transmission.

Key features include:

- **Sequence numbers** – Each frame carries a unique sequence number for identification
- **Timer mechanism** – Sender starts a timer after transmitting each frame
- **Retransmission** – If ACK is not received before timer expires, frame is retransmitted
- **Error detection** – Protocol can detect corrupted frames and acknowledgments

* Data frame can be lost
* Ack can be lost

Lost data frames:
This means that sender will never get an ack.
Sender can implement timer for each frame sent. If timer expires retransmit.

But what if the data frame arrived but the ack was lost?

Using above method, sender will retransmit frame that receiver has already received.
Result: *Duplicate frame at sender.*

So receiver needs some way of distinguishing duplicates.
Answer: Use sequence numbers.

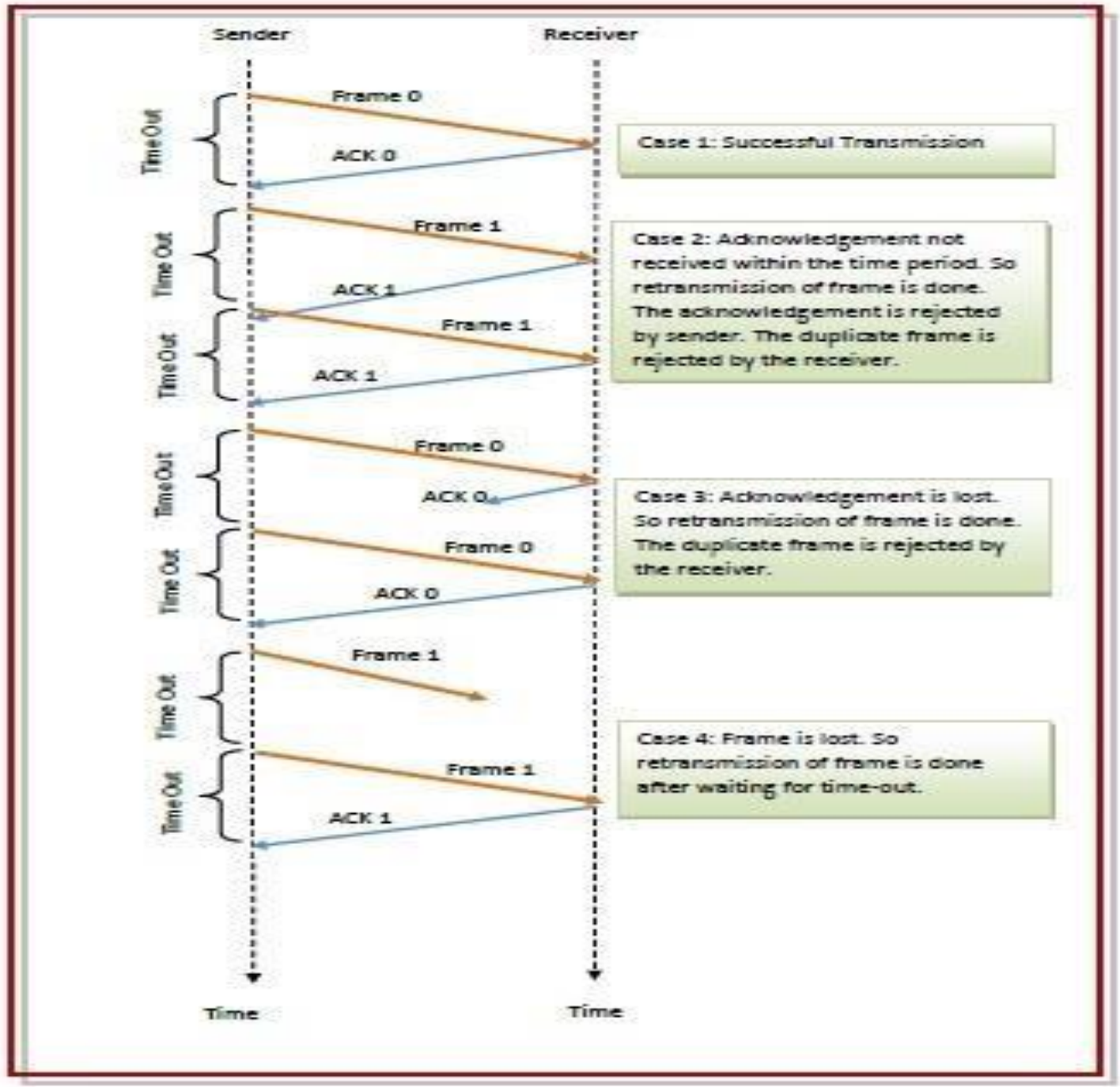
What should be the field size of these sequence numbers ?

•**Sender Side** −

At the sender site, a field is added to the frame to hold a sequence number. If data is available, the data link layer makes a frame with the certain sequence number and sends it. The sender then waits for arrival of acknowledgment for a certain amount of time. If it receives a positive acknowledgment for the frame with that sequence number within the stipulated time, it sends the frame with next sequence number. Otherwise, it resends the same frame.

•**Receiver Side** −

The receiver also keeps a sequence number of the frames expected for arrival. When a frame arrives, the receiver processes it and checks whether it is valid or not. If it is valid and its sequence number matches the sequence number of the expected frame, it extracts the data and delivers it to the network layer. It then sends an acknowledgement for that frame back to the sender along with its sequence number.


Sender
Receiver
Time Out
Frame 0
ACK 0
Case 1: Successful Transmission
Time Out
Frame 1
ACK 1
Time Out
Frame 1
ACK 1
Case 2: Acknowledgement not received within the time period. So retransmission of frame is done. The acknowledgement is rejected by sender. The duplicate frame is rejected by the receiver.
Time Out
Frame 0
ACK 0
Time Out
Frame 0
ACK 0
Case 3: Acknowledgement is lost. So retransmission of frame is done. The duplicate frame is rejected by the receiver.
Time Out
Frame 1
Time Out
Frame 1
ACK 1
Case 4: Frame is lost. So retransmission of frame is done after waiting for time-out.
Time
Time

Sliding window protocol:

- Sliding window protocols are data link layer protocols for reliable and sequential delivery of data frames. The sliding window is also used in Transmission Control Protocol.

- In this protocol, **multiple frames can be sent by a sender at a time before receiving an acknowledgment from the receiver.**

- The **term sliding window refers to the imaginary boxes to hold frames**.

- Sliding window method is also known as windowing.

How data is transmitted**.** **It follows 2 parameters**

1) **sequence number.** (given to each outbound i.e. sending frame) (range 0 to 2^n-1).

2) **sliding window**. (imaginary boxes i.e. buffers temporary storage)
Transmitter and receiver side to increase efficiency

- The size of the receiving window is the maximum number of frames that the receiver can accept at a time.

- It determines the maximum number of frames that the sender can send before receiving acknowledgment.

Working Principle:

- In these protocols, the sender has a buffer called the sending window and the receiver has buffer called the receiving window.

- The size of the sending window determines the sequence number of the outbound frames.

- If the sequence number of the frames is an **n-**bit field, then the range of sequence numbers that can be assigned is **0 to 2^n−1**. Consequently, the size of the sending window is 2^n−1.

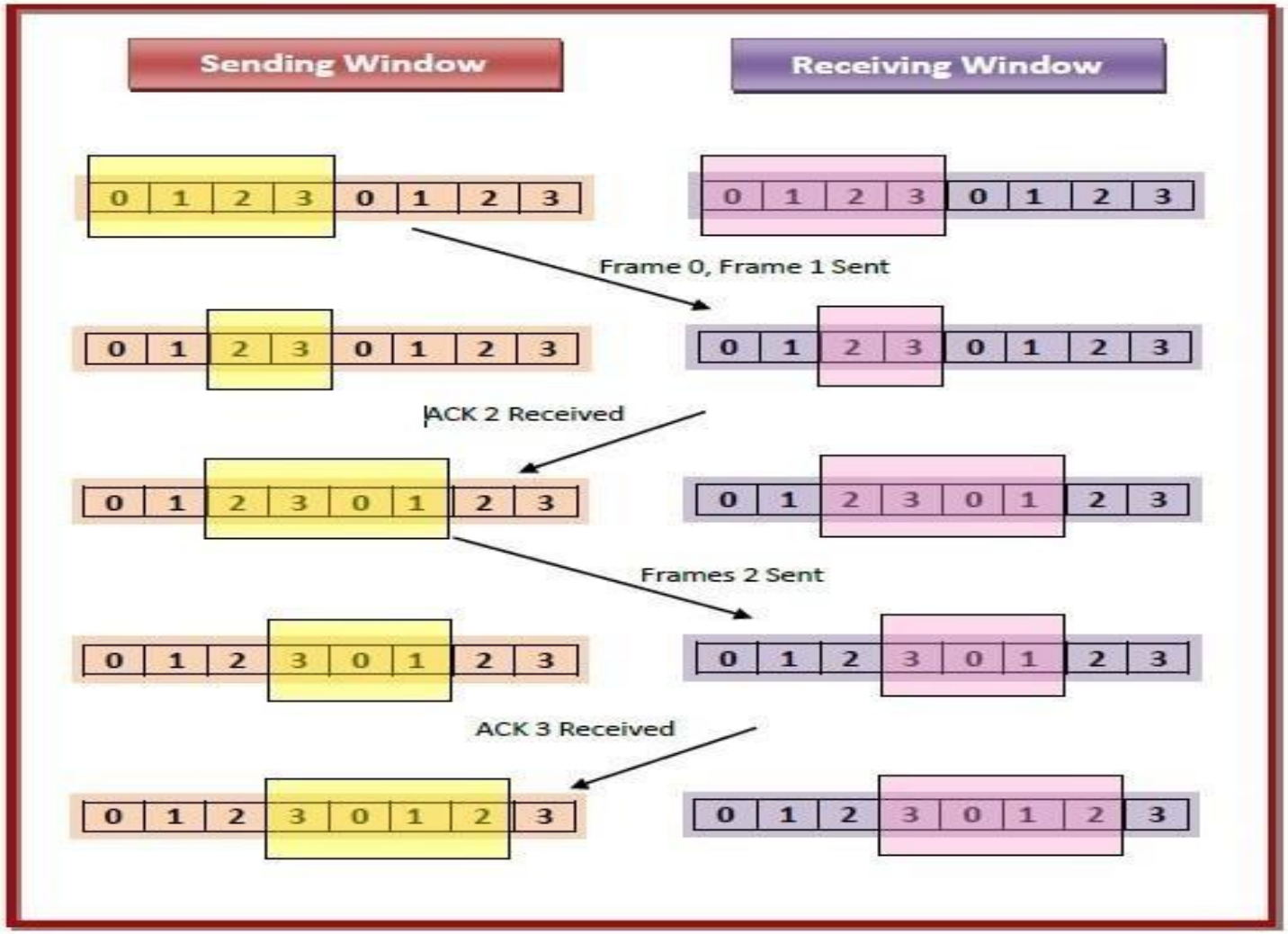
- **For example:**

  if the sending window size is 4, then the sequence numbers will be 0, 1, 2, 3, 0, 1, 2, 3, 0, 1, and so

  on. The number of bits in the sequence

Example:

Suppose that we have sender window and receiver window each of size 4. So the sequence numbering of both the windows will be 0,1,2,3,0,1,2 and so on. The following diagram shows the positions of the windows after sending the frames and receiving acknowledgments.



One-Bit Sliding window protocol:

- This is '**Simplex protocol' with sequence number and Acknowledgement frame indicating the sequence number of the next frame expected.**

- **In this sliding window protocol the maximum window size is 1. such a protocol uses stop-and-wait since the sender transmits a frame and waits for its Acknowledgement before sending next frame.**

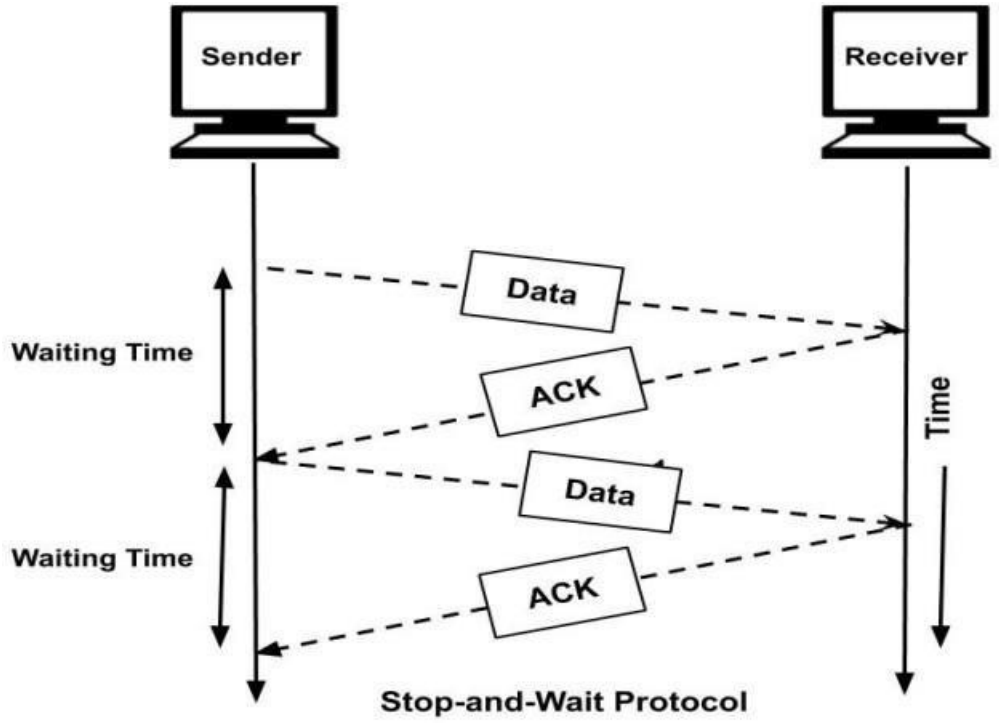
- One bit sliding window protocol is also called as Stop-and-wait ARQ (automatic repeat request).

- It performs Error control and flow control mechanisms.

  a) **Normal Operation.**
  b) **Frame lost operation.**
  c) **Acknowledgment lost operation.**
  d) **Acknowledgement delay operation.**

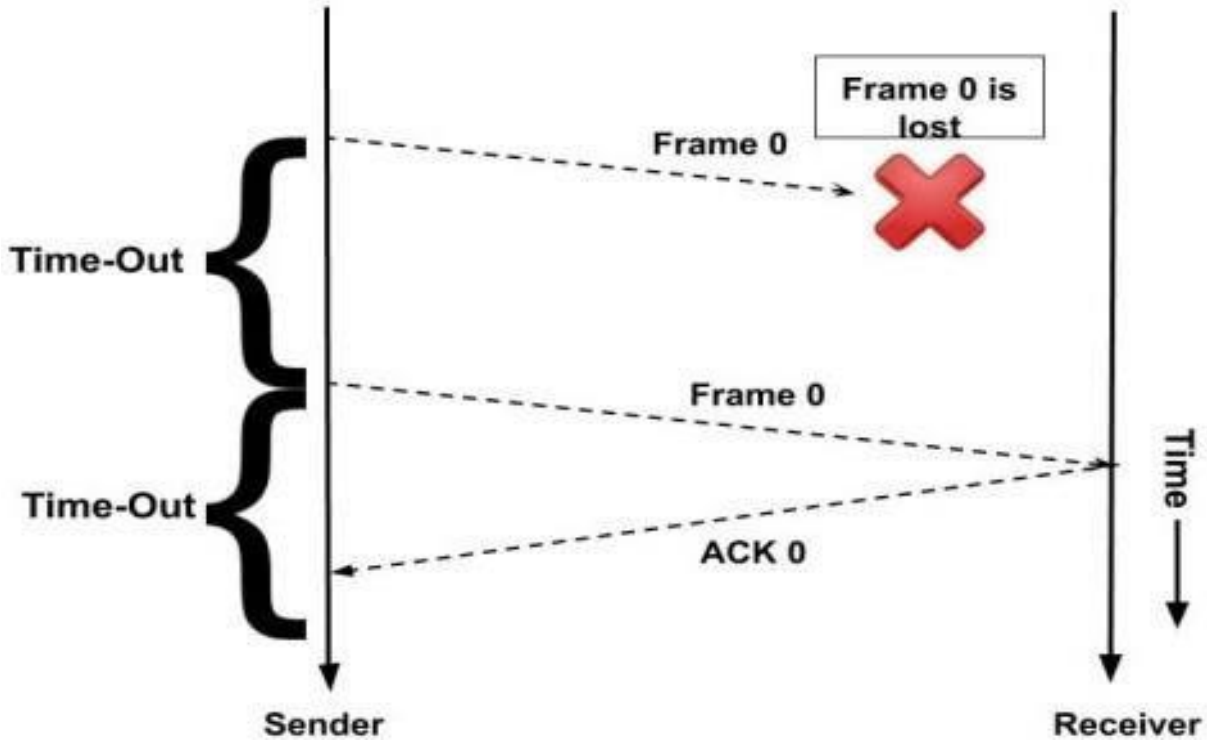
a) Normal Operation:

It is the simplest flow control method. In this, the sender will send one frame at a time to the receiver. The sender will **stop and wait** for the acknowledgment from the receiver. This time(i.e. the time between message sending and acknowledgement receiving) is the waiting time for the sender and the sender is totally idle during this time. When the sender gets the acknowledgment(ACK), then it will send the next data packet to the receiver and wait for the acknowledgment again and this process will continue as long as the sender has the data to send.



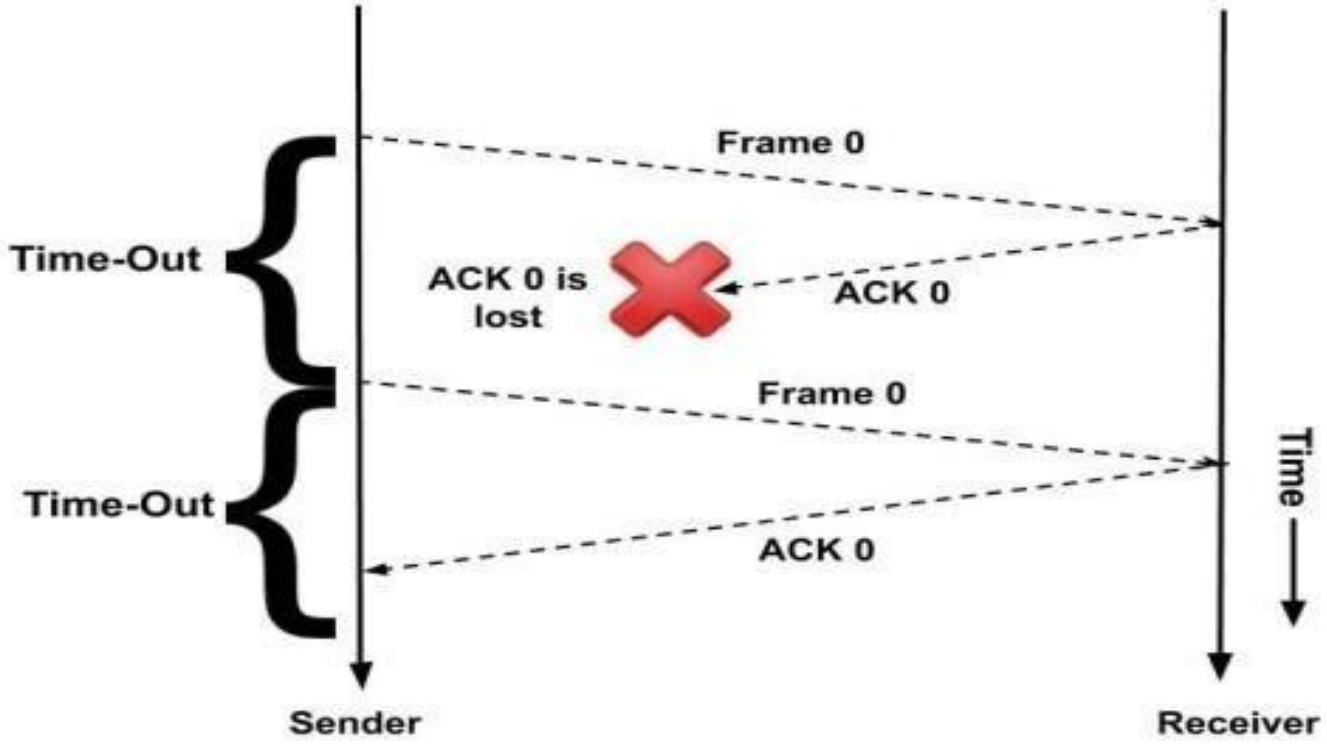
b) Frame loss operation :

Suppose if any frame sent is not received by the receiver and is lost. So the receiver will not send any acknowledgment as it has not received any frame. Also, the sender will not send the next frame as it will wait for the acknowledgment for the previous frame which it had sent. So a deadlock situation arises here. To avoid any such situation there is a **time-out** timer. The sender waits for this fixed amount of time for the acknowledgment and if the acknowledgment is not received then it will send the frame again.

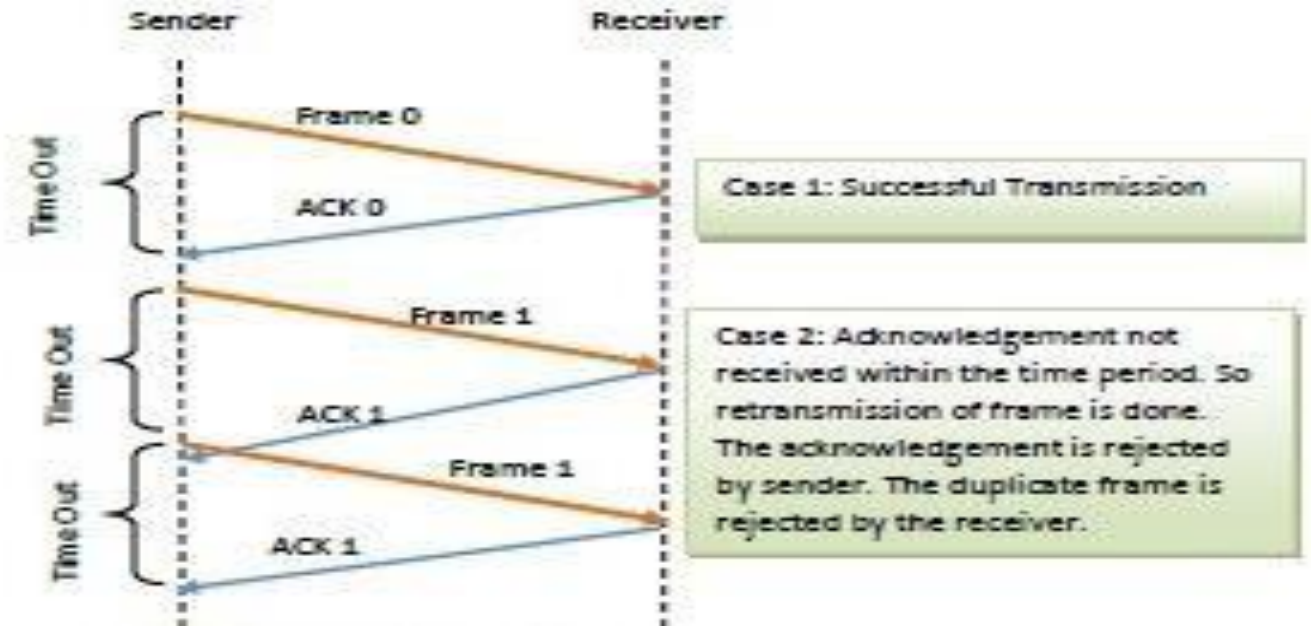


c) Acknowledgement loss :

Consider a situation where the receiver has received the data and sent the acknowledgment but the ACK is lost. So, again the sender might wait till infinite time if there is no system of **time-out** timer. So, in this case also, the time-out timer will be used and the sender will wait for a fixed amount of time for the acknowledgment and then send the frame again if the acknowledgement is not received.



d) Delay Acknowledgement :



Types of Sliding Window Protocols:

The Sliding Window ARQ (Automatic Repeat request) protocols are of two categories −



- **Go – Back – *N* ARQ:**

- Go – Back – *N* ARQ provides for sending multiple frames before receiving the acknowledgment for the first frame. It uses the concept of sliding window, and so is also called sliding window protocol. The frames are sequentially numbered and a finite number of frames are sent. If the acknowledgment of a frame is not received within the time period, all frames starting from that frame are retransmitted.

- **Selective Repeat ARQ**
- This protocol also provides for sending multiple frames before receiving the acknowledgment for the first frame. However, here only the erroneous or lost frames are retransmitted, while the good frames are received and buffered.

Go – Back – *N* ARQ:

- **N indicates number of frames to be send.**

- **The difference b/w stop & wait protocol and Go-back-N protocol Stop &wait sends only one frame at a time but in this protocol sends several frames Before receiving Acknowledgment.**

- **It is Error control and Detect error is Data link layer.**

Go – Back – *N* ARQ provides for sending multiple frames before receiving the acknowledgment for the first frame.

It uses the concept of sliding window, and so is also called sliding window protocol.

The frames are sequentially numbered and a finite number of frames are sent.

If the acknowledgment of a frame is not received within the time period, all frames starting from that frame are retransmitted.

The Receiver Receives Damaged frame the receiver sends Negative Acknowledgement with Sequence number of Damaged frame i.e. N values ,Again it sends all frame from damaged frame to until last frame once again.

Damaged Frame:

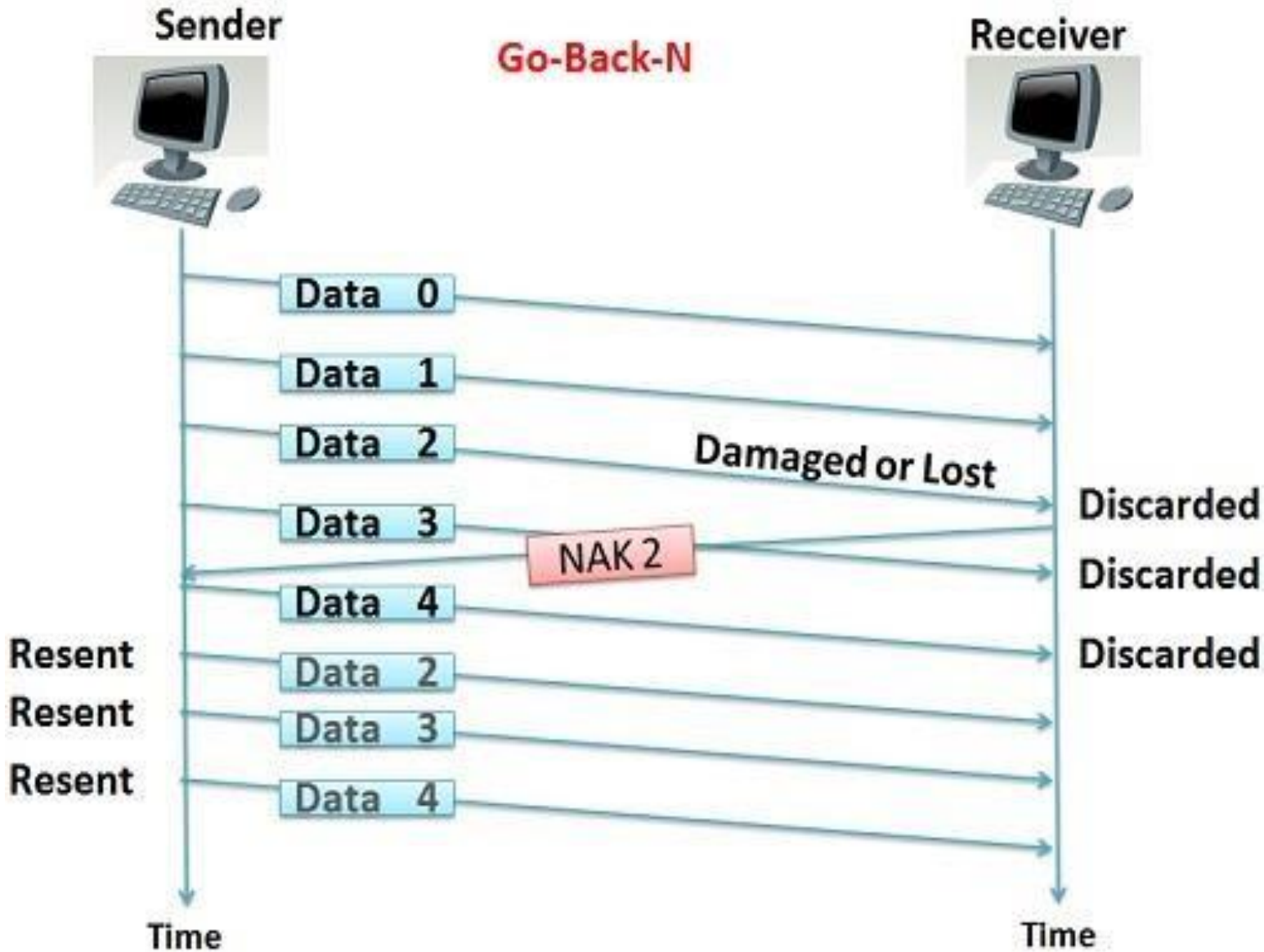
- **Receiver** sends Negative Acknowledgement along with Frame Sequence number.
- After Negative Acknowledgement receives Discard all frames.

Loss Frame:

- Checks the number on each frame.
- Verifies the sequence number i.e. easily identifies which frame is missing.

Loss Acknowledgement:

- **If** the sender does not receives Acknowledgement within a time so agian it transmits .


Sender
Go-Back-N
Receiver
Data 0
Data 1
Data 2
Damaged or Lost
Discarded
Data 3
NAK 2
Discarded
Data 4
Discarded
Resent
Data 2
Resent
Data 3
Resent
Data 4
Time
Time

Additional features of Go-Back-N ARQ

1. Sequence numbers : Sequence numbers of transmitted frame are maintained in the header of each frame. If k is the number of bits for sequence number, then the numbering can range from 0 to $2^k - 1$ e.g. for k = 3. Sequence numbers are 0 to 7 $(2^3 - 1)$.

2. Sender sliding window : Window is a set of frames in buffer waiting for acknowledgement. This window keeps on sliding in forward direction. The window size is fixed. As the ACK is received, the respective frame goes out of window and new frame to sent come into window. Fig. 2.4.7 illustrates sliding of window for window size = 7.

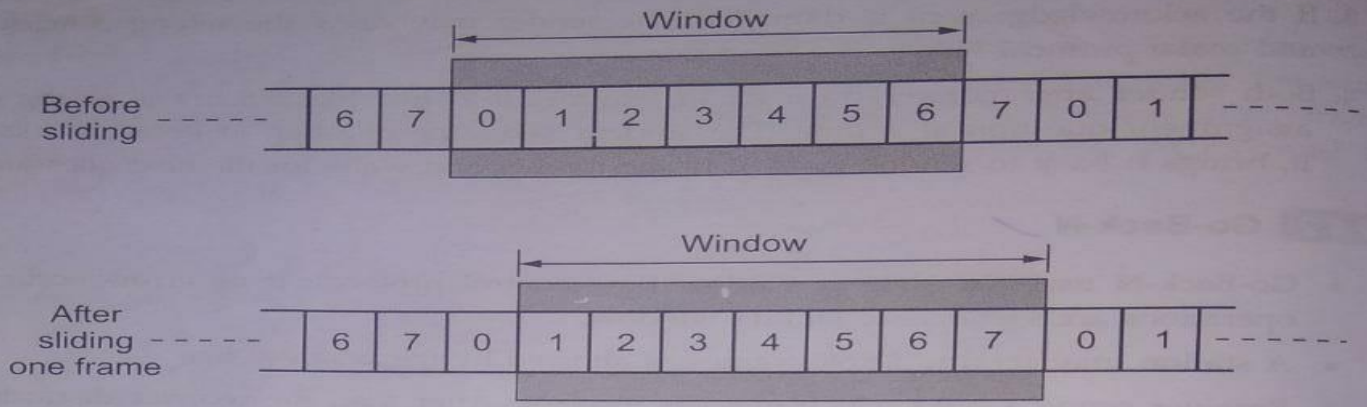


Fig. 2.4.7 Sender sliding window

3. Receiver sliding window : In the receiver side the size of the window is always one. The receiver is expecting to arrive frames in specific sequence. Any other frame received which is out of order is discarded. The receiver slides over after receiving the expected frame. Fig. 2.4.8 shows receiver sliding window.

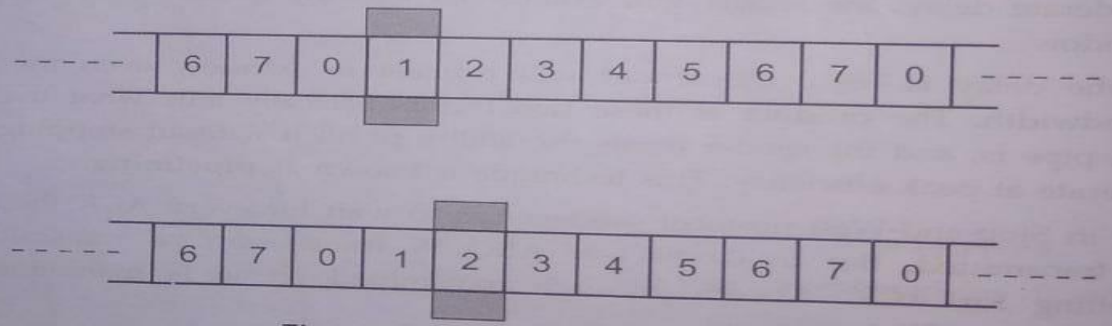


Fig. 2.4.8 Receiver sliding window

4. Control variables :

a) Sender variables

- The sender deals with three different variables.

$S \rightarrow$ sequence number of recently sent frame.

$S_F \rightarrow$ sequence number of first frame in window.

$S_L \rightarrow$ sequence number of last frame in window.

$\therefore$ Window size $W = S_L - S_F + 1$

e.g. In previous feature, $W = 7 - 0 + 1 = 8$

b) Receiver variable

- The receiver deals with one variable only.
  R → sequence no. of frame expected.

  If the number matches, then the frame is accepted otherwise not.

5. Timers

- The sender has a timer for each transmitted frame. The receiver don't have any timer.

6. Acknowledgement

- The receiver responds for frames arriving safely by positive acknowledgements. For damaged or lost frames receiver does not reply, the sender has to retransmit it when timer of that frame elapsed.
- The receiver may acknowledge once for several frames.

7. Resending of frames

- If the timer for any frame expires, the sender has to resend that frame and the subsequent frames also, hence the protocol is called Go-Back-N ARQ.

Operation

a) Normal operation

- The sender sends frames and update the control variables i.e. S_F, S, S_L and receiver updates variable R. Fig. 2.4.9 shows normal operation.

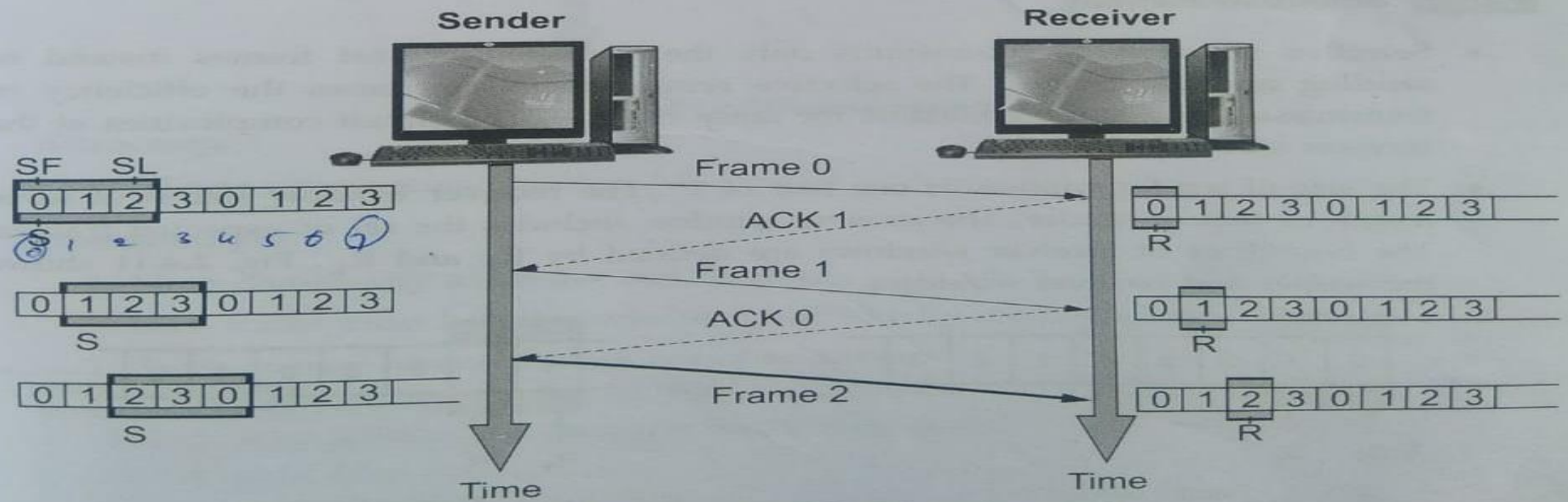


Fig. 2.4.9 Go-Back-N ARQ, Normal operation

b) Damaged or lost frame

- Suppose frame 2 is damaged or lost and if receiver receives frame 3, it will be discarded since it is expecting frame 2. Sender retransmits frame 2 and frame 3. Fig. 2.4.10 shows this process.

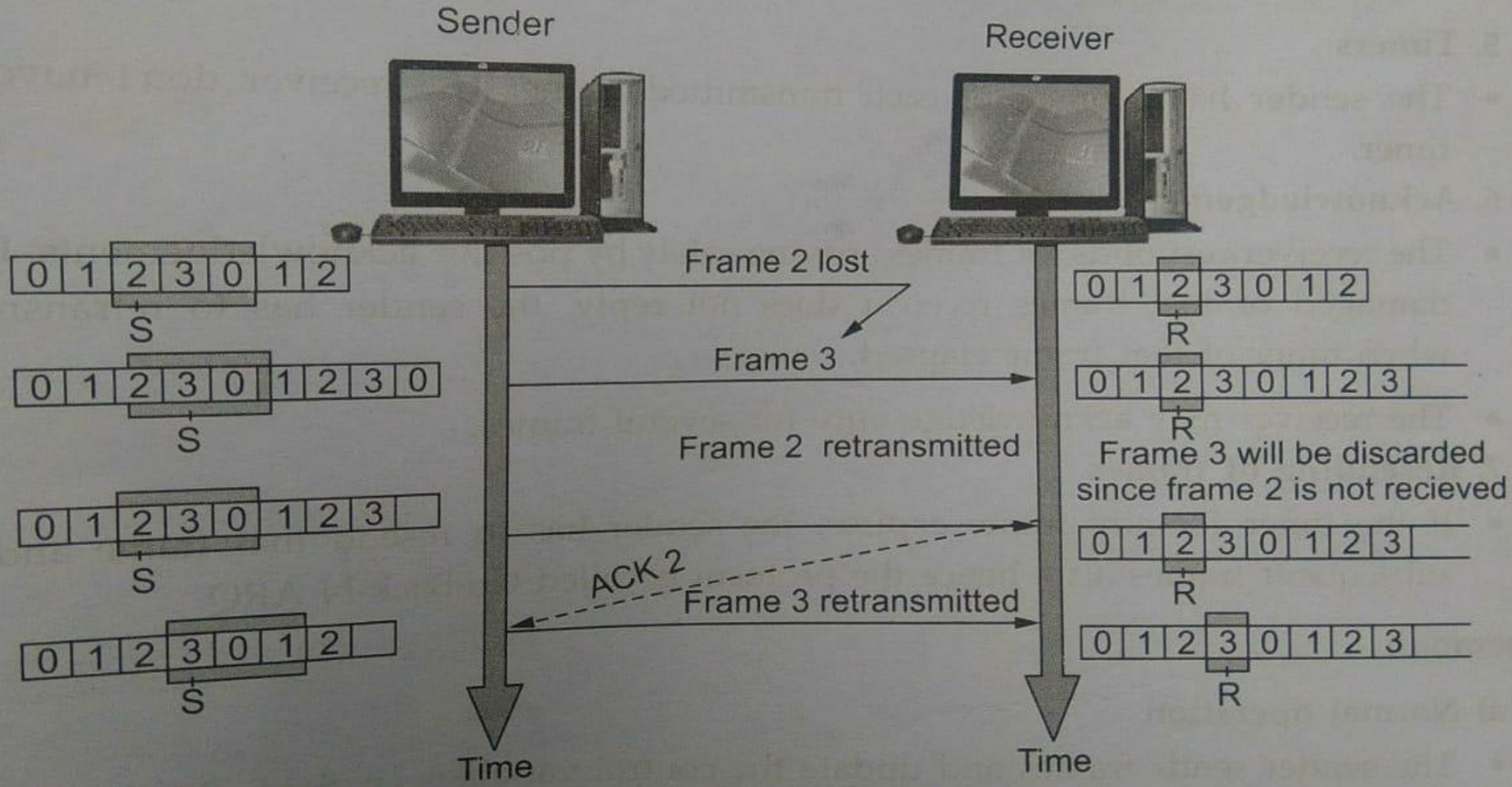
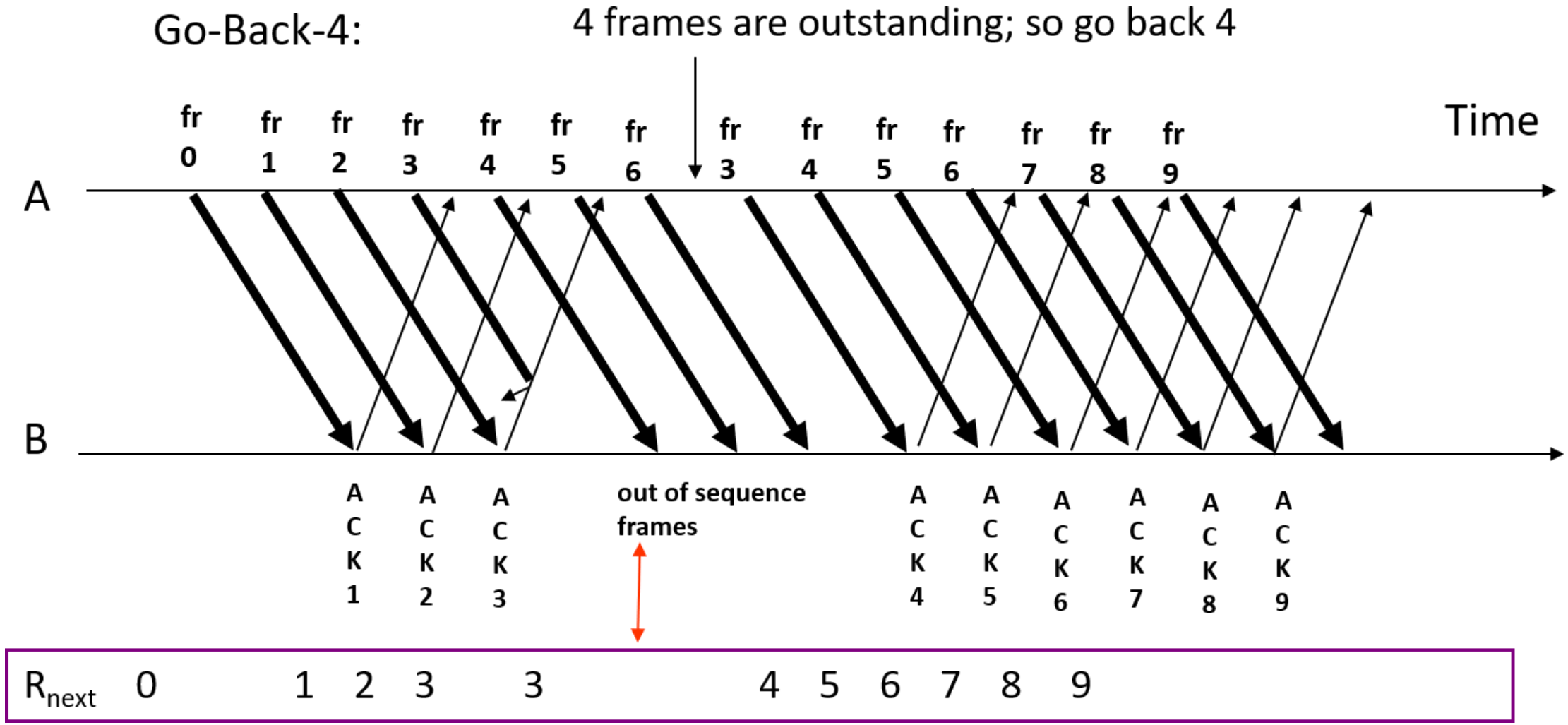
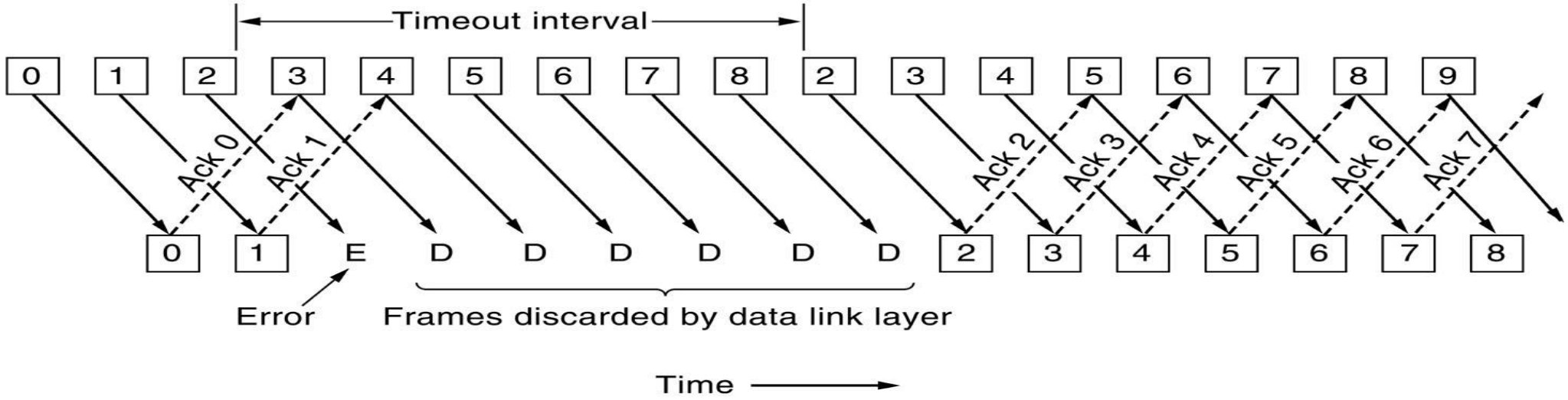
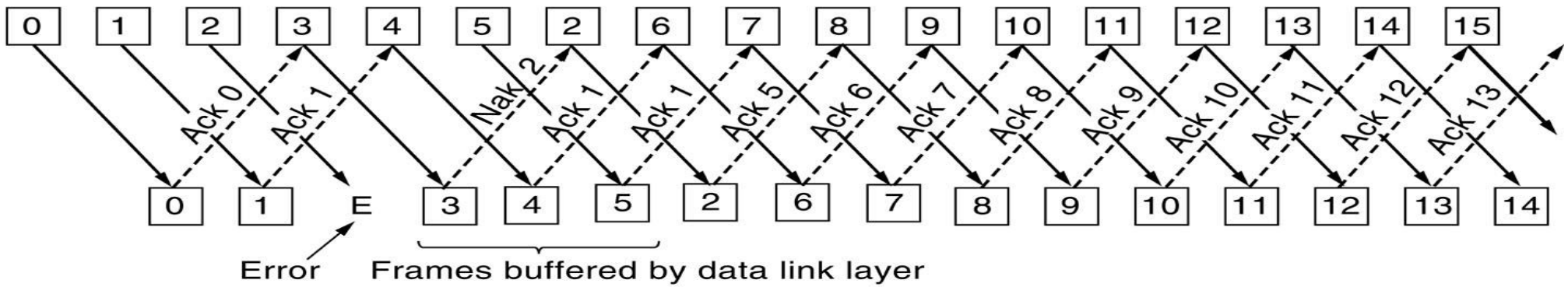


Fig. 2.4.10 Go-Back-N, Damaged frame


Go-Back-4:
4 frames are outstanding; so go back 4
Time
A
B
fr 0
fr 1
fr 2
fr 3
fr 4
fr 5
fr 6
fr 3
fr 4
fr 5
fr 6
fr 7
fr 8
fr 9
ACK1
ACK2
ACK3
out of sequence frames
ACK4
ACK5
ACK6
ACK7
ACK8
ACK9
Rnext 0 1 2 3 3 4 5 6 7 8 9


Timeout interval
0 1 2 3 4 5 6 7 8 2 3 4 5 6 7 8 9
Ack 0
Ack 1
Ack 2
Ack 3
Ack 4
Ack 5
Ack 6
Ack 7
0 1 E D D D D D D 2 3 4 5 6 7 8
Error
Frames discarded by data link layer
Time

(a)


0 1 2 3 4 5 2 6 7 8 9 10 11 12 13 14 15
Ack 0
Ack 1
Nak 2
Ack 1
Ack 1
Ack 5
Ack 6
Ack 7
Ack 8
Ack 9
Ack 10
Ack 11
Ack 12
Ack 13
0 1 E 3 4 5 2 6 7 8 9 10 11 12 13 14
Error
Frames buffered by data link layer

(b)

Selective Repeat ARQ:

•This protocol also provides for sending multiple frames before receiving the acknowledgment for the first frame.

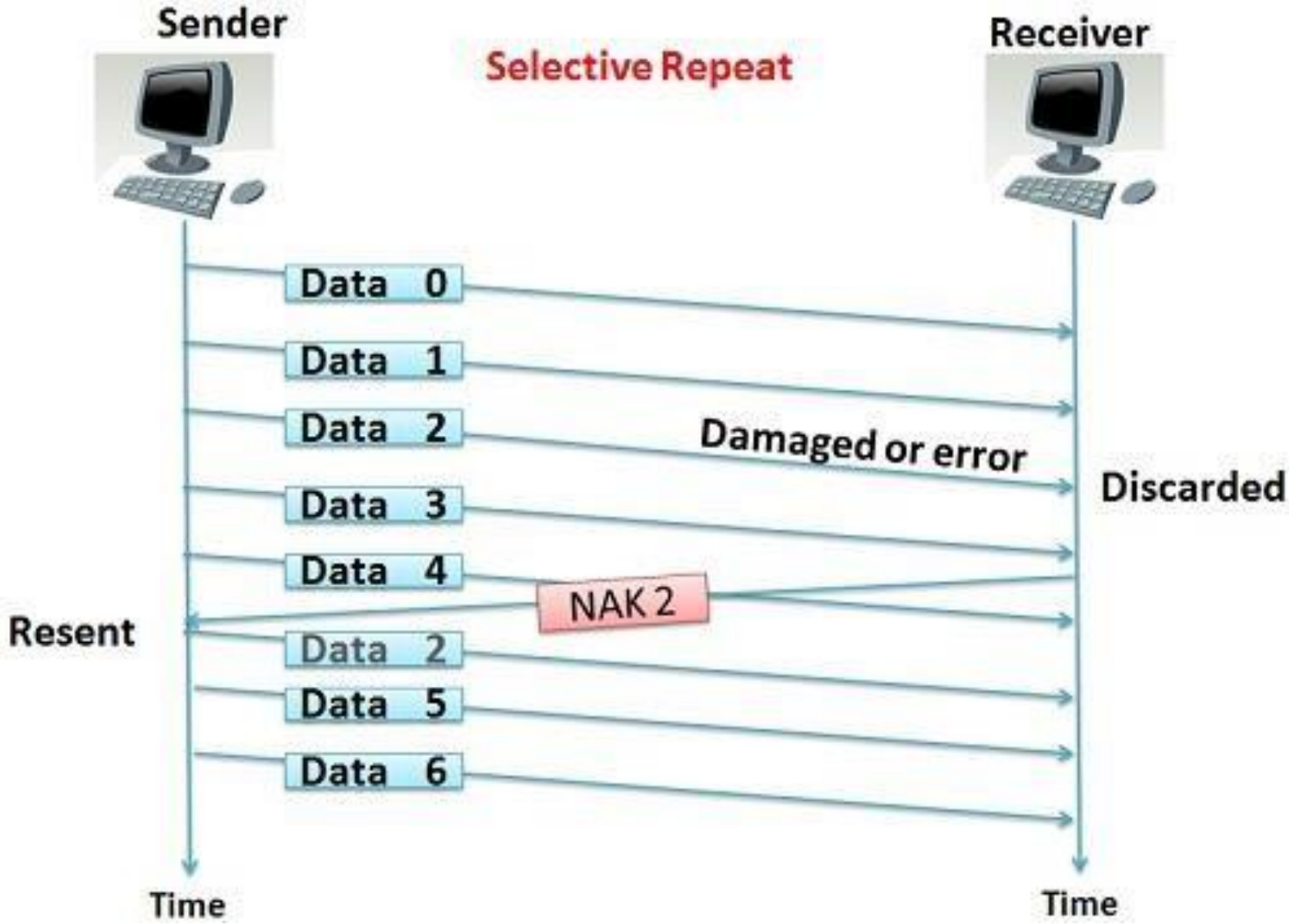
•However, here only the lost frames are retransmitted, while the good frames are received and buffered.

•It Retransmits only Damaged or Loss frame only.

•The receiver is capable of Sorting the frame proper sequence manner.

•The sender must be capable of searching frame for which NAK has been received.

•It requires Less Window size as compare to Go-Back-N Protocol.


Sender
Receiver
Selective Repeat
Data 0
Data 1
Data 2
Damaged or error
Discarded
Data 3
Data 4
NAK 2
Resent
Data 2
Data 5
Data 6
Time
Time

Operation

- In sequential transmission of frame 0, 1, 2, 3, suppose frame 2 is lost and the next frame 3 is already received then receiver sends NAK 2 frame to sender. Then sender retransmits frame 2 only. Fig. 2.4.12 shows operation of selective repeat ARQ.

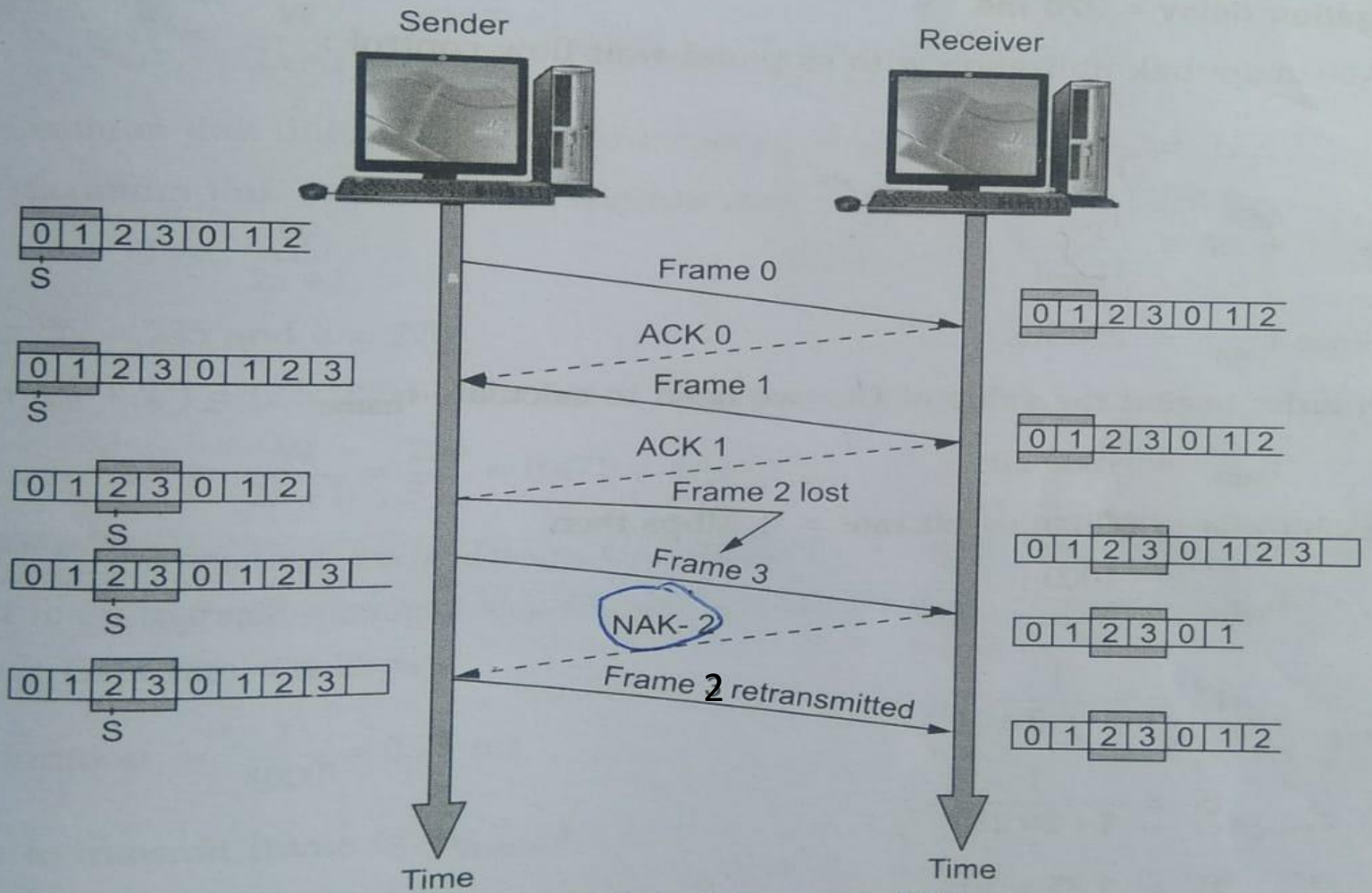


Fig. 2.4.12 Selective repeat ARQ

S.NO	GO-BACK-N PROTOCOL	SELECTIVE REPEAT PROTOCOL
1.	In Go-Back-N Protocol, if the sent frame are find suspected then all the frames are re-transmitted from the lost packet to the last packet transmitted.	In selective Repeat protocol, only those frames are re-transmitted which are found suspected.
2.	Sender window size of Go-Back-N Protocol is N.	Sender window size of selective Repeat protocol is also N.
3.	Receiver window size of Go-Back-N Protocol is 1.	Receiver window size of selective Repeat protocol is N.
4.	Go-Back-N Protocol is less complex.	Selective Repeat protocol is more complex.
5.	In Go-Back-N Protocol, neither sender nor at receiver need sorting.	In selective Repeat protocol, receiver side needs sorting to sort the frames.
6.	In Go-Back-N Protocol, Out-of-Order packets are NOT Accepted (discarded) and the entire window is re-transmitted.	In selective Repeat protocol, Out-of-Order packets are Accepted.
7.	In Go-Back-N Protocol, if Receives receives a corrupt packet, then also, the entire window is re-transmitted.	In selective Repeat protocol, if Receives receives a corrupt packet, it immediately sends a negative acknowledgement and hence only the selective packet is retransmitted.

Error Detection and Error correction:
Hamming Code:

- Hamming code is a set of error-correction codes that can be used to **detect and correct the errors** that can occur when the data is moved or stored from the sender to the receiver.

- In this method System to identify where Error is occur and Rectify that Error.

- It is **technique developed by R.W. Hamming for error correction**.

Redundant bits –

- Redundant bits are extra binary bits that are generated and added to the information-carrying bits of data transfer to ensure that no bits were lost during the data transfer.

The number of redundant bits can be calculated using the following formula:

$$2^r \geq m + r + 1$$

where, r = redundant bit, m = data bit

Suppose the number of data bits is **7**,
then the number of redundant bits can be calculated using: = **$2^4 \geq 7 + 4 + 1$**
Thus, the number of redundant bits= **4**

Data (M) -----→ Any Length

(parity bits) Redundant Bits (r) ----------→ Find Length of r by using below equation.

$$2^r \geq m + r + 1$$

where, r = redundant bit, m = data bit

Sender ----→ send the message **M + r**

(here ' r ' placed powers of 2)

R1 ---→ 2^0
R2 ---→ 2^1
R3 ---→ 2^2
R4 ---→ 2^4

Depends on the length of 'r' we have to place these redundant bits.

Ex :Redundant bits (r) =4

P8	P4	P2	P1
2^4	2^2	2^1	2^0

Parity bits –

A parity bit is a bit appended to a data of binary bits to ensure that the total number of 1's in the data is even or odd. Parity bits are used for error detection.

There are two types of parity bits:

Even parity bit:

In the case of even parity, for a given set of bits, the number of 1's are counted. If that count is odd, the parity bit value is set to 1, making the total count of occurrences of 1's an even number. If the total number of 1's in a given set of bits is already even, the parity bit's value is 0.

Odd Parity bit :

In the case of odd parity, for a given set of bits, the number of 1's are counted. If that count is even, the parity bit value is set to 1, making the total count of occurrences of 1's an odd number. If the total number of 1's in a given set of bits is already odd, the parity bit's value is 0.

General Algorithm of Hamming code –

The Hamming Code is simply the use of extra parity bits to allow the identification of an error.

1. Write the bit positions starting from 1 in binary form (1, 10, 11, 100, etc i.e 1,2,3,4……).
2. All the bit positions that are a power of 2 are marked as parity bits (1, 2, 4, 8, etc).
3. All the other bit positions are marked as data bits.
4. Each data bit is included in a unique set of parity bits, as determined its bit position in binary form.

a. Parity bit 1 covers all the bits positions whose binary representation includes a 1 in the least significant position (1, 3, 5, 7, 9, 11, etc).

b. Parity bit 2 covers all the bits positions whose binary representation includes a 1 in the second position from
the least significant bit (2, 3, 6, 7, 10, 11, etc).

c. Parity bit 4 covers all the bits positions whose binary representation includes a 1 in the third position from
the least significant bit (4–7, 12–15, 20–23, etc).

d. Parity bit 8 covers all the bits positions whose binary representation includes a 1 in the fourth position from
the least significant bit bits (8–15, 24–31, 40–47, etc).

e. In general, each parity bit covers all bits where the bitwise AND of the parity position and the bit position is
non-zero.

5. Since we check for even parity set a parity bit to 1 if the total number of ones in the positions it checks is
odd.

6. Set a parity bit to 0 if the total number of ones in the positions it checks is even.

Position	R8	R4	R2	R1
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10	1	0	1	0
11	1	0	1	1

R1 -> 1,3,5,7,9,11
R2 -> 2,3,6,7,10,11
R3 -> 4,5,6,7
R4 -> 8,9,10,11

Haming Code

Data (M) → Any Length

(parity bits) Redundant bits (r) → length of 'r' depends on

$$2^r \geq m + r + 1$$

Sender Sends → $m + r$ Information.

↓

Placed at powers of '2'.

$r_1 \rightarrow 2^0$

$r_2 \rightarrow 2^1$

$r_3 \rightarrow 2^2$

$r_4 \rightarrow 2^3$

(r_1 placed at 2^0, $r_1 \rightarrow 2^1$ ----)

$r = 4$ (parity bits)

P_8	P_4	P_2	P_1
↓	↓	↓	↓
2^3	2^2	2^1	2^0

→ positions of parity bits.

Either we can follow "Even parity" (or) "ODD parity".

Even Parity (Ex):

Data (M) = 1001 ← 4 bits.

(Redundant) parity bits (r) = $2^r \geq m + r + 1$ satisfies the condition.

$2^1 \geq 4 + 1 + 1$ | $2^2 \geq 4 + 2 + 1$ | $2^3 \geq 4 + 3 + 1$

$2 \geq 6$ ✗ | $4 \geq 7$ ✗ | $8 \geq 8$ ✓

satisfies $r = 3$ ✓

$\therefore (r) = 3$ bits

P_4	P_2	P_1

2020-9-20 12:08

$(r) = 3$ bits

P_4	P_2	P_1

→ powers of 2 $(2^0, 2^1, 2^2)$

1 2 4 positions

Length of Data $= M + r$

$= 4 + 3$

$= (7)$ bits total.

Now calculate powers of '2'

7	6	5	4	3	2	1
1	0	0	P_4	1	P_2	P_1

(i.e) 1 0 0 P_4 1 P_2 P_1

then sender Identify the P_1, P_2, P_4 bits.

Then Sender Identify the P_1, P_2, P_4 bits.

Now Calculate All possible Binary Representation of 3 bits

	P_4	P_2	P_1
1 →	0	0	1
2 →	0	1	0
3 →	0	1	1
4 →	1	0	0
5 →	1	0	1
6 →	1	1	0
7 →	1	1	1

(Binary Representation)

→ Now find position of 1's in Binary Representation (Even parity)

P_1 → 1, 3, 5, 7 → P_1 1 0 1 → $P_1 = 0$

P_2 → 2, 3, 6, 7 → P_2 1 0 1 → $P_2 = 0$

P_4 → 4, 5, 6, 7 → P_4 0 0 1 → $P_4 = 1$

↓

Values of particular position in Data frame

P_4 101 → 0 (Corrupted)

2020-09-27

P_1 101 → (0)101 → Even parity

P_2 101 → (0)101 → Even parity

P_4 001 → (1)001 → Even parity.

Now Here $P_1=0$, $P_2=0$, $P_4=1$ ⟹ Replace these value in original Data.

7	6	5	4	3	2	1
1	0	0	(1)	1	(0)	(0)
			P_4		P_2	P_1

∴ These All Calculation done at "Sender" Level.

→ Now the frame "1001100" is send to the Receiver.

Receiver Receive Corrupted data: ← 1 (1) 0 1 1 0 0

7	6	5	4	3	2	1
1	(1)	0	1	1	0	0

(With Error)

2020-9-20 12:09

2020-9-20 12:

Receiver Receives Corrupted data: ← 1 1 0 1 1 0 0
(7 6 5 4 3 2 1)
(With Error)

(find Position)

$P_1 \rightarrow 1, 3, 5, 7 \rightarrow 0101 \rightarrow P_1 = 0$

$P_2 \rightarrow 2, 3, 6, 7 \rightarrow 0111 \rightarrow P_2 = 1$

$P_4 \rightarrow 4, 5, 6, 7 \rightarrow 1011 \rightarrow P_4 = 1$

Consider Even parity

if we get
P_1, P_2, P_4 all 0's
then it is Original Data

Now find Error

P_4	P_2	P_1
1	1	0

→ Decimal (6) So 6th position Error.

0

7 6 5 4 3 2 1
1 1 0 1 1 0 0

6th position Now Chage '1' to '0'

Medium Access sub layer:

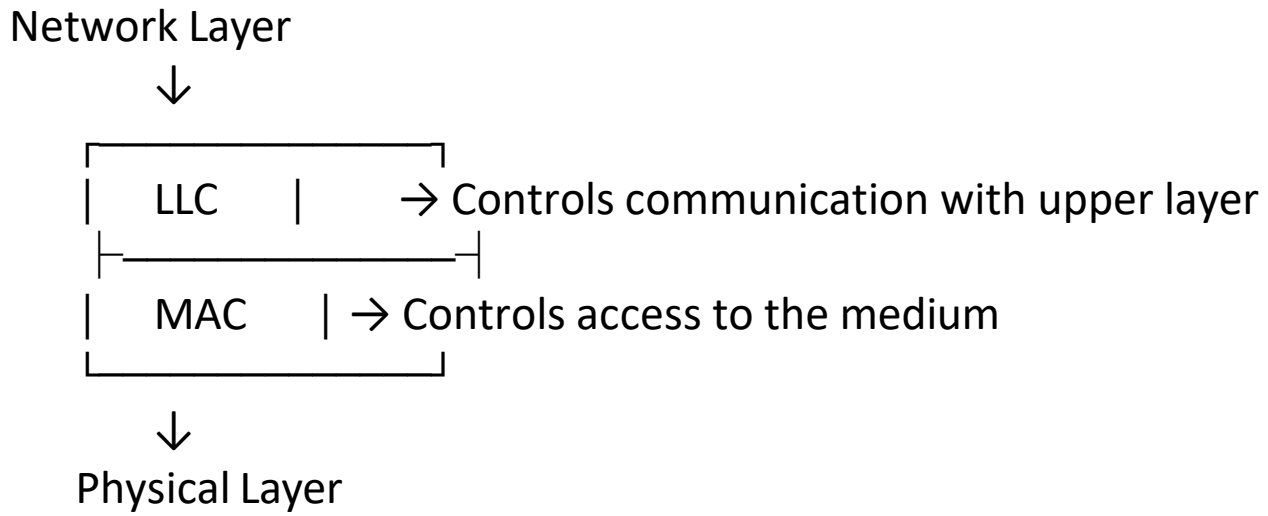
MAC Layer in the OSI Model:

The Open System Interconnections (OSI) model is a layered networking framework that conceptualizes **how communications should be done between heterogeneous systems**.

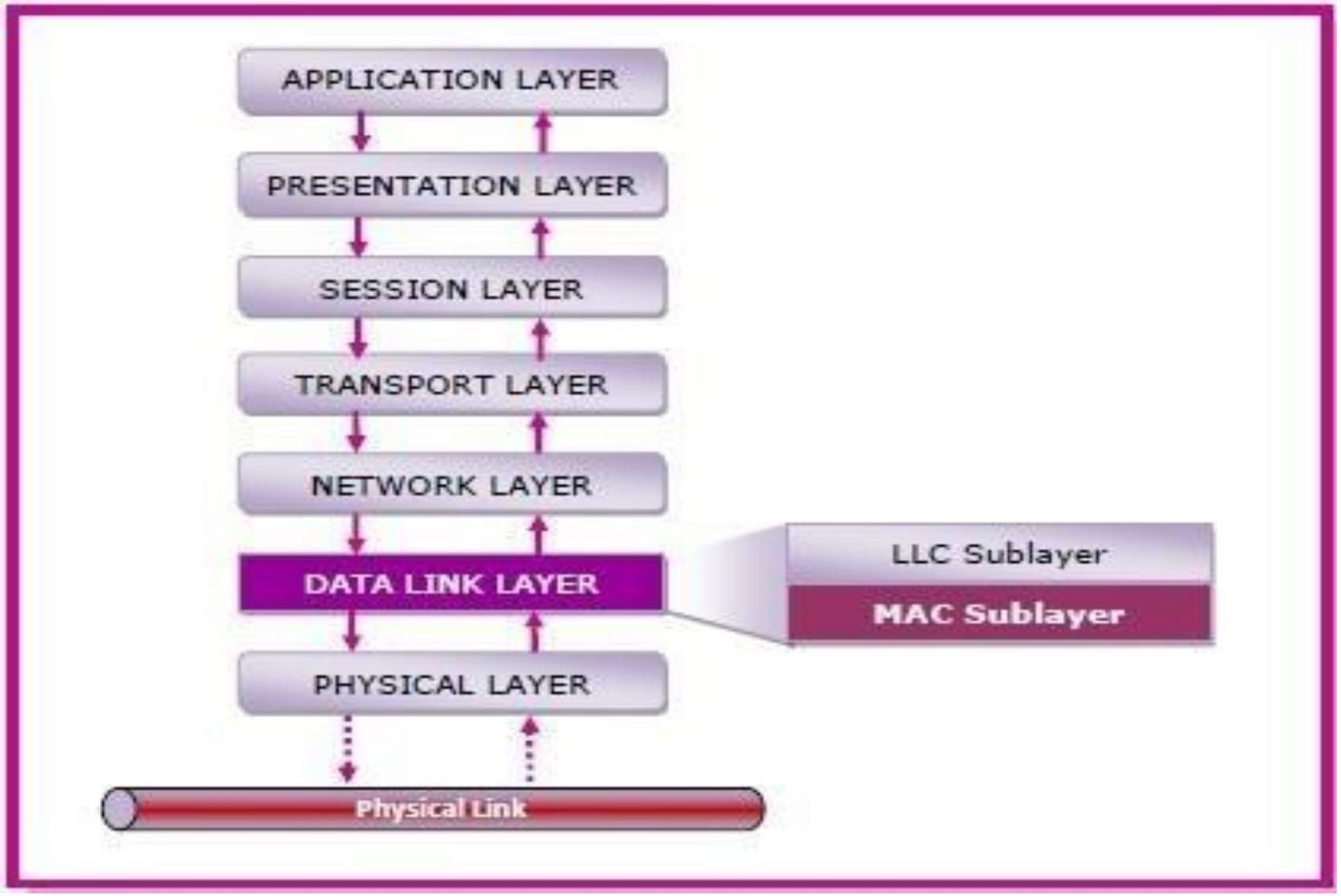
The data link layer is the second lowest layer.

It is divided into two sublayers:

* The logical link control (LLC) sublayer
* The medium access control (MAC) sublayer



The following diagram depicts the position of the MAC layer −



Functions of MAC Layer:

In **Computer Networks (CN)**, the **MAC (Medium Access Control) sublayer** is the lower part of the Data Link Layer.
Functions of the MAC Sublayer

Framing – Helps organize transmitted data into frames.

Physical addressing – Uses **MAC addresses** to identify devices on a network.

Media access control – Determines **which device can access the shared transmission medium** and when.

Collision handling – Helps detect or avoid collisions when multiple devices transmit simultaneously (e.g., CSMA/CD, CSMA/CA).

Error detection – Uses mechanisms such as **CRC** to **detect errors** in frames.

Frame delivery – **Controls the transmission of frames** between devices over the physical medium.

Access coordination – **Manages and sharing of the communication channel among multiple devices**.

The MAC sublayer mainly controls **how devices access the shared medium and how frames are delivered using MAC addresses**.

Channel Allocation Problem:

- **A single channel is divided and shared among multiple users.**

  → The available communication bandwidth is divided so that several users can transmit data.

- **The number of users may vary over time.**

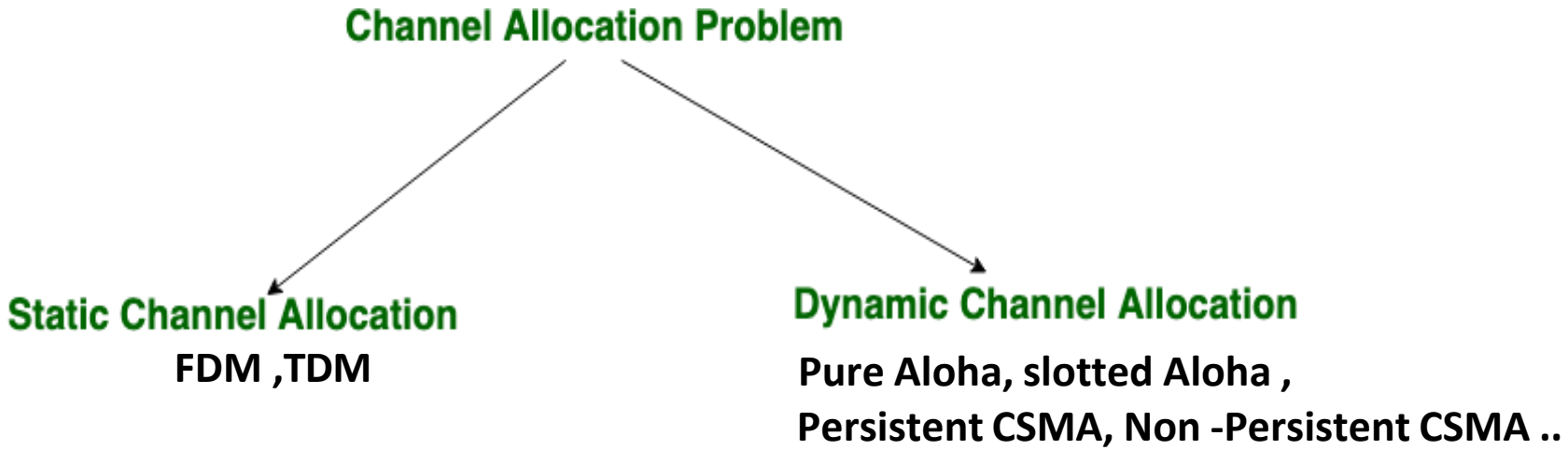
  → Users may join or leave the network, so the channel requirements can change.

- **If there are N users, the channel can be divided into N equal sub-channels.**

  → Each sub-channel gets a fixed portion of the total bandwidth.

- **Each user gets one sub-channel.**

  → A user can use its assigned portion to send and receive data without interfering with other users.



1. Static Channel Allocation in LANs and MANs:

Static channel allocation means the communication channel is divided into **fixed portions** and assigned to users **in advance**. The allocation does not change according to the current traffic.

1. FDM – Frequency Division Multiplexing

The total channel bandwidth is divided into **different frequency bands**.

Each user is assigned a **fixed frequency band**.

Users can transmit **simultaneously** using different frequencies.

Example: If there are 4 users, the available frequency range can be divided into 4 bands.

Advantage: Users can transmit continuously without waiting for a time slot.

2. TDM – Time Division Multiplexing

The channel is divided into **fixed time slots**.

Each user is assigned a specific time slot.

Users transmit **one after another** in their assigned slots.

Example: With 4 users, the channel can have 4 time slots: User 1 → User 2 → User 3 → User 4.

Advantage: The same frequency/channel can be shared efficiently.

2. Dynamic Channel Allocation:

Dynamic channel allocation is a method in which the communication channel is **allocated to users only when they need to transmit data**. The allocation changes according to the current traffic and number of active users.

- **Main Features**
- Channel is **not permanently assigned** to any user.
- Users get access to the channel **when they have data to send**.
- Channel resources are shared among **active users**.
- Allocation changes according to the **network traffic**.
- It provides better **bandwidth utilization** than static allocation.
- It reduces the wastage of channel capacity.
- It is useful when the **number of users and traffic vary frequently**.
- Multiple users may try to access the channel at the same time, which can cause **collisions**.

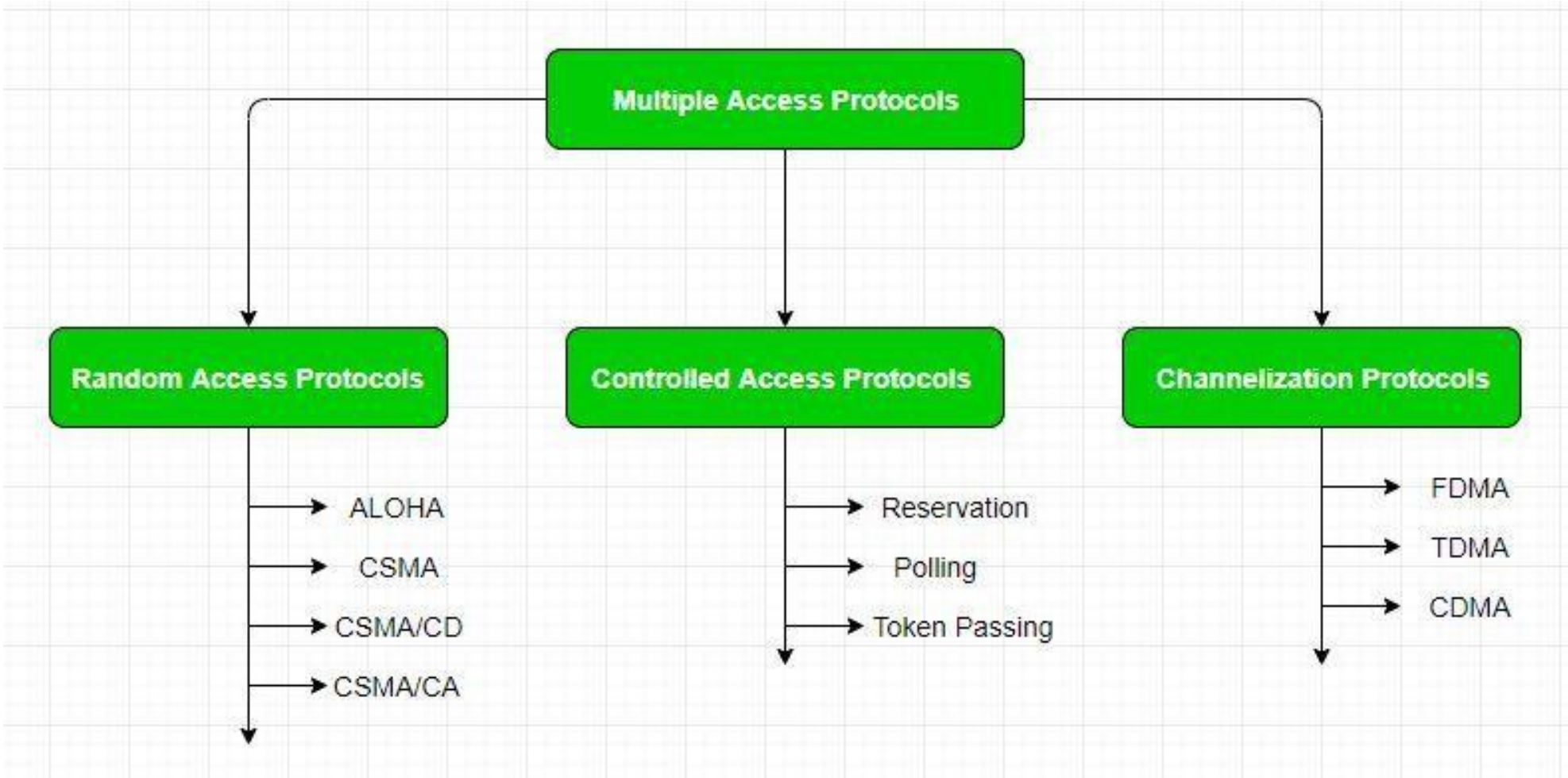
Multiple Access Protocols:

•If there is a **dedicated link between the sender and the receiver** then data link control layer is sufficient.

• however if there is **no dedicated link present then multiple stations can access the channel simultaneously**.

•Hence **multiple access protocols are required to decrease collision and avoid crosstalk.**

For example: In a classroom full of students, when a teacher asks a question and all the students (or stations) start answering simultaneously (send data at same time) then a lot of collisions is created(data overlap or data lost) then it is the job of the teacher (multiple access protocols) to manage the students and make them answer one at a time.


Multiple Access Protocols
Random Access Protocols
Controlled Access Protocols
Channelization Protocols
ALOHA
CSMA
CSMA/CD
CSMA/CA
Reservation
Polling
Token Passing
FDMA
TDMA
CDMA

1. Random Access Protocol:

In this, **all stations have same superiority that is no station has more priority than another station. Any station can send data depending on medium's state(idle or busy).**

It has two features:

1. There is no fixed time for sending data.
2. There is no fixed sequence of stations sending data.

The Random access protocols are further subdivided as:

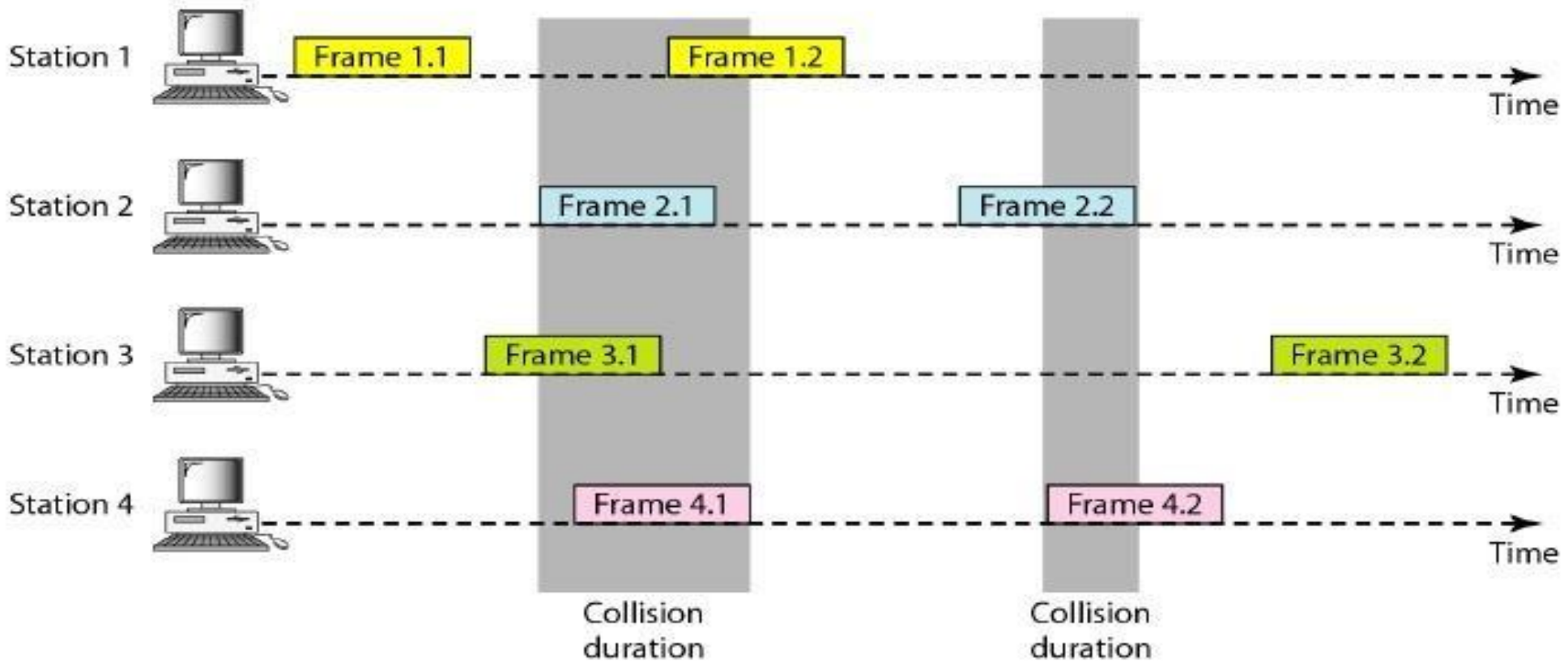
(a) ALOHA –

- It was **designed for wireless LAN but is also applicable for shared medium.**
- In this, **multiple stations can transmit data at the same time and can hence lead to collision and data being garbled (confuse).**

Pure Aloha:

- **Pure Aloha is an un-slotted, decentralized, and simple to implement a protocol**.
- In pure ALOHA, the **stations simply transmit frames whenever they want data to send.**
- It **does not check whether the channel is busy or not before transmitting**.
- In case, **two or more stations transmit simultaneously, the collision occurs and frames are destroyed.**
- **Whenever any station transmits a frame**, it **expects the acknowledgment from the receiver**.
- If **it is not received within a specified time**, the **station assumes that the frame or acknowledgment has been destroyed.**
- Then, the **station waits for a random amount of time and sends the frame again**.
- This **randomness helps in avoiding more collisions.**
- This **scheme works well in small networks** where the load is not much. **But in largely loaded networks, this scheme fails poorly**. This led to the **development of Slotted Aloha.**
- Since different stations wait for different amount of time, the probability of further collision decreases.

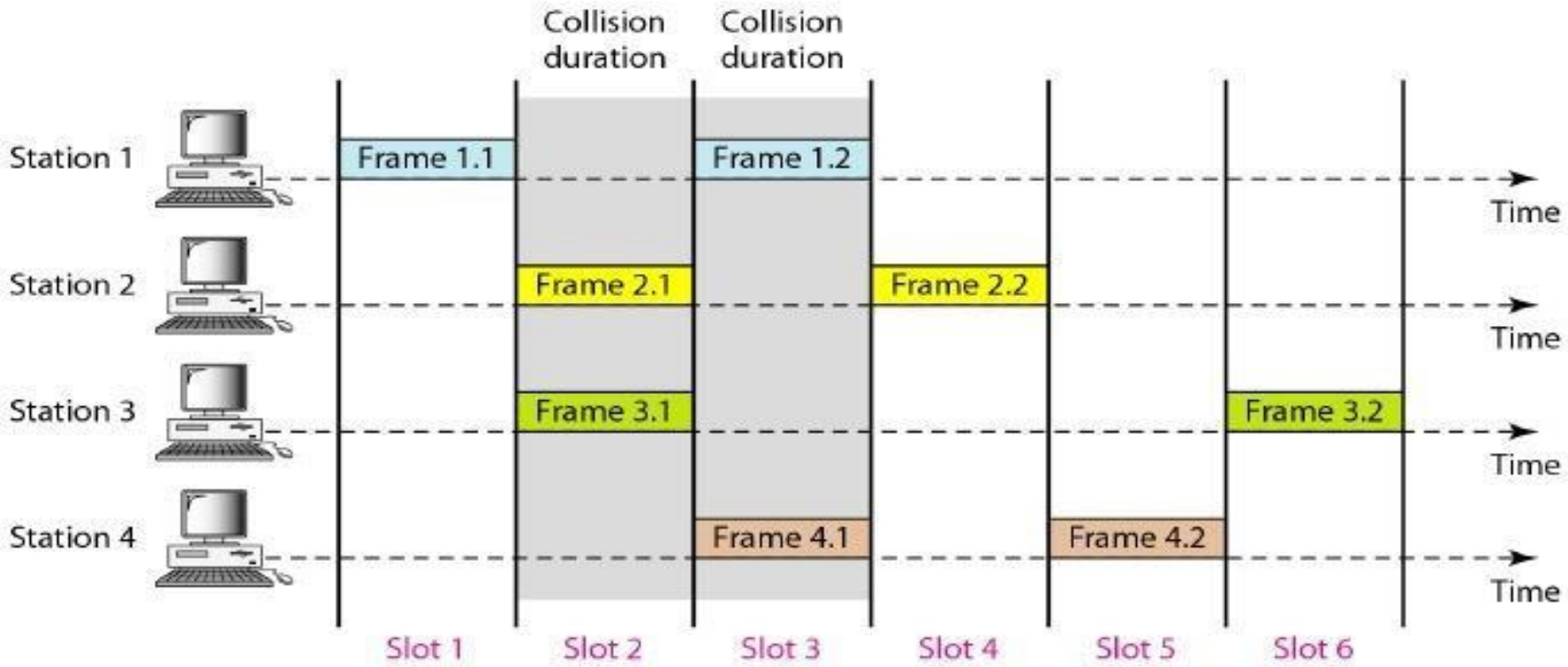
Figure 12.3 *Frames in a pure ALOHA network*

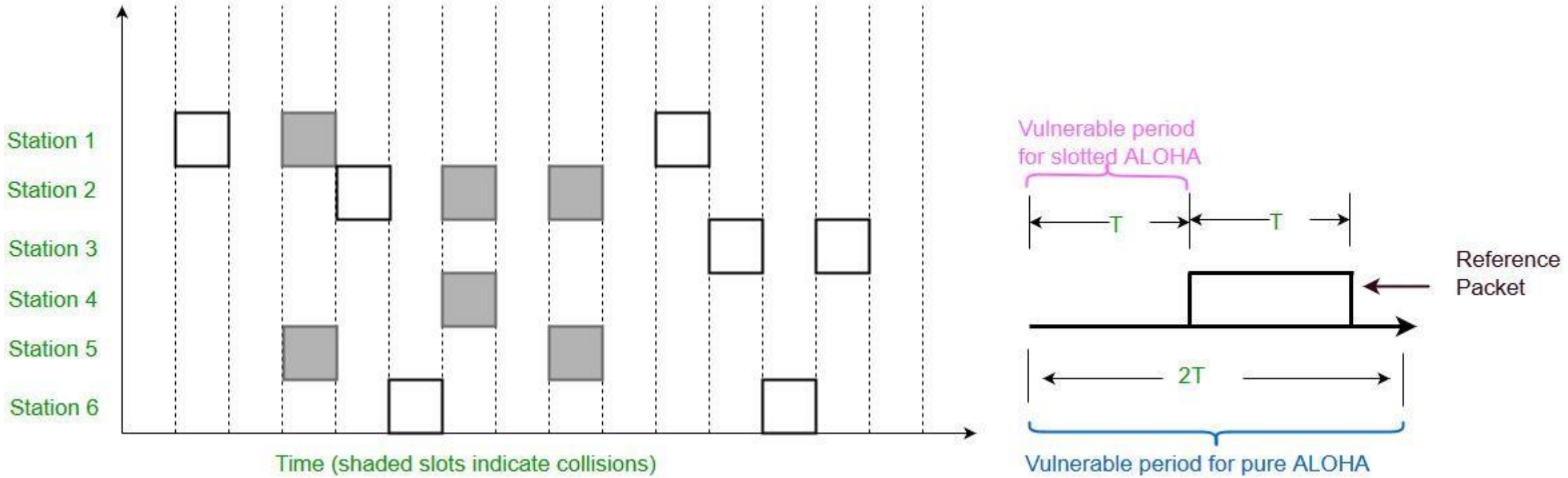


Slotted Aloha:

- It is **similar to pure aloha**, **except that we divide time into slots and sending of data is allowed only at the beginning of these slots.**
- **If a station misses out the allowed time**, it **must wait for the next slot**.
- This is quite similar to Pure Aloha, **differing only in the way transmissions take place**.
- **Instead of transmitting** right **at demand time**, the **sender waits for some time**.
- In slotted ALOHA, the time of the **shared channel is divided into discrete intervals called Slots.**
- The stations are eligible to **send a frame only at the beginning of the slot** and **only one frame per slot is sent**.
- **If any station is not able to place the frame onto the channel at the beginning of the slot**, it has to **wait until the beginning of the next time slot.**
- There is **still a possibility of collision if two stations try to send at the beginning of the same time slot.**
- **But still the number of collisions** that can possibly take place is **reduced by a large margin and the performance becomes much well compared to Pure Aloha.**
- This **reduces the probability of collision.**

Figure 12.6 *Frames in a slotted ALOHA network*





Collision is possible for only the current slot. Therefore, Vulnerable Time is Tt.

`Vulnerable Time = Frame transmission time. V=Tt`

The average amount of transmission-attempts for 2 consecutive frame

Efficiency of Slotted ALOHA:

S*slotted* = G * e^-G

Maximum Efficiency: (**S**slotted)*max* = 1 * e^-1 = 1/e = 0.368

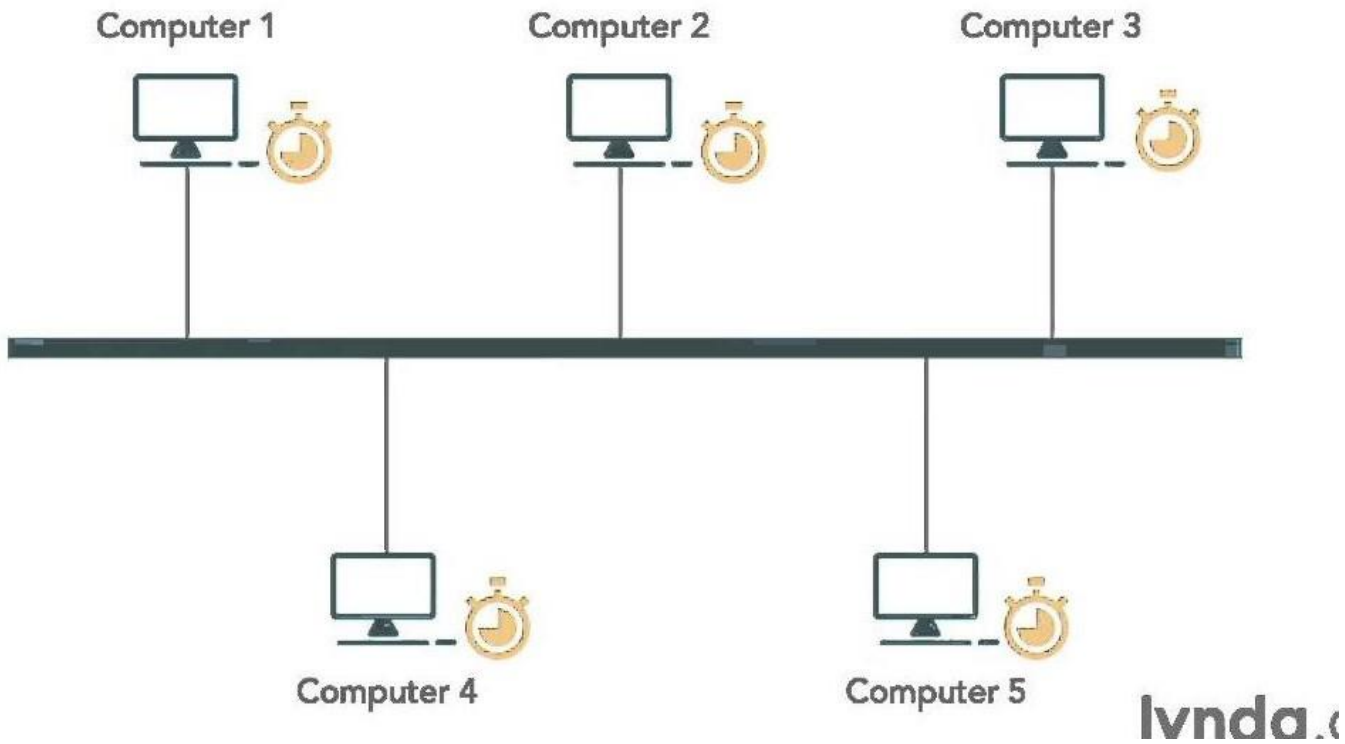
	Key	Pure Aloha	Slotted Aloha
1	Time Slot	In Pure Aloha, any station can transmit data at any time.	In Slotted Aloha, any station can transmit data only at beginning of any time slot.
2	Time	Frames are transmitted in arbitrary time	Time is divided up to discrete slot , the frame is sent at the starting of a slot.
3	Time	In Pure Aloha, time is continuous and is not globally synchronized.	In Slotted Aloha, time is discrete and is globally synchronized.
4	Vulnerable time	Vulnerable time is 2 times of frame transmission time. Ie. Vulnerable time = 2 x Tt.	Vulnerable time is one half of pure aloha. Vulnerable time = Tt.
5	Probability	Probability of successful transmission of data packet = $G \times e^{-2G}$	Probability of successful transmission of data packet = $G \times e^{-G}$
6	Maximum efficiency	Maximum efficiency = 18.4%.	Maximum efficiency = 36.8%.
7	Number of collisions	Does to reduces the number of collisions.	Slotted Aloha reduces the number of collisions to half thus doubles the efficiency.

(b) CSMA – (Carrier Sense Multiple Access)

- **Carrier** means channel or medium .
- To **minimize the chance of collision** and therefore to **Increase the performance**.
- Principle of CSMA **“ Sense(identify) before transmit the data or listen before transmits the data”.**
- **Without sensing the channel state** to transmits the data through the channel collision may occur.
- **Carrier busy = Transmission is taking place.**
- **Carrier idle = No transmission currently taking place.**
- Carrier Sense Multiple Access ensures fewer collisions as the station is required to first sense the medium (for idle or busy) before transmitting data.
- If it is idle then it sends data, otherwise it waits till the channel becomes idle.
- However there is still chance of collision in CSMA due to **propagation delay.**

- For **example**, if station A wants to send data, it will first sense the medium . If it finds the channel idle, it will start sending data. However, by the time the first bit of data is transmitted (delayed due to propagation delay) from station A, if station B requests to send data and senses the medium it will also find it idle and will also send data. This will result in collision of data from station A and B.

- The possibility of Collison still exists because of propagation delay : a station may sense the medium and find it is idle because the first bit sent by another station has not yet been received.



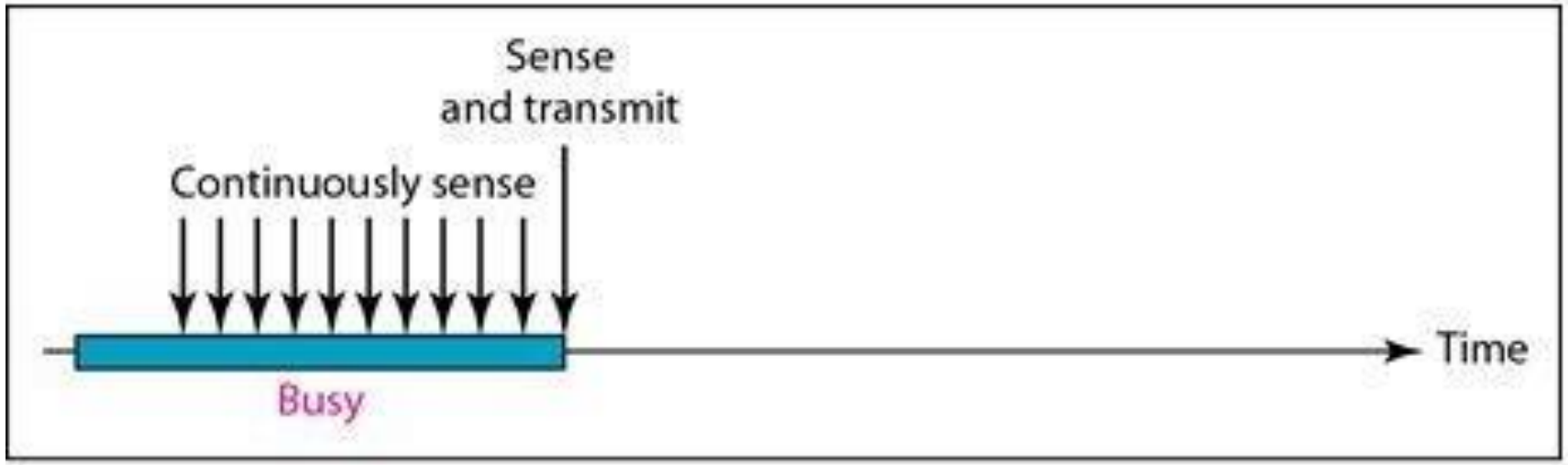
Types of CSMA:

- **1-Persistent CSMA.**
- **P-Persistent CSMA.**
- **Non-Persistent CSMA.**
- **O-Persistent CSMA.**

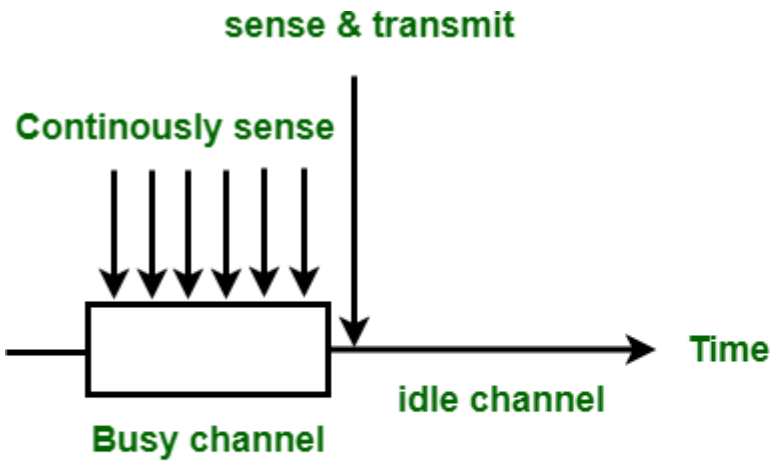
CSMA access modes-

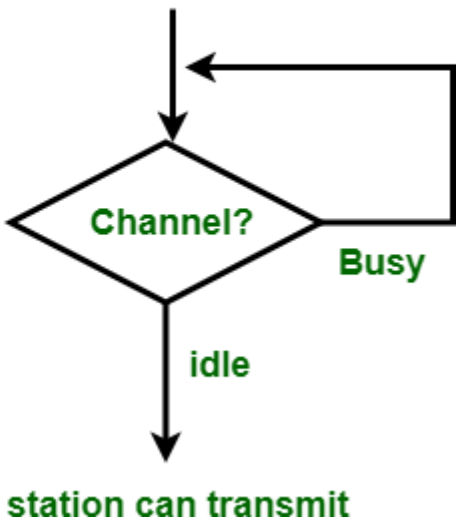
1-persistent CSMA:

- Before sending the data the station **first listens** to the channel to see if anyone else to transmitting the data at this moment.
- IF **Listen Primitive tell if the channel is busy or idle** if it is idle to transmits the data other wise wait until free .
- If the **channel is idle** the station transmits the frame.
- If **channel is busy** then it **sense the transmission medium continuously until it becomes idle.**
- **Since the station** transmits the frame with the probability of 1 when the carrier or channel is idle , this scheme of CSMA is called 1- Persistent CSMA.
- The propagation delay has an important effect on the performance of the protocol.
- The node senses the channel, if idle it sends the data, otherwise it continuously keeps on checking the medium for being idle and transmits unconditionally(with 1 probability) as soon as the channel gets idle.
- The **longer propagation delay** . The importance is **more effected because performance of protocol is decreases.**



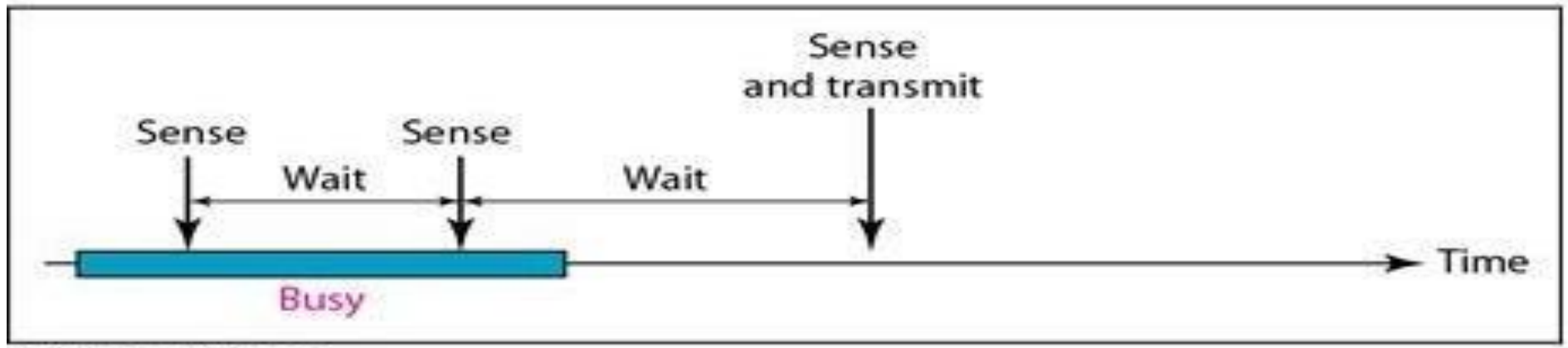
a. 1-persistent





Non-Persistent CSMA:

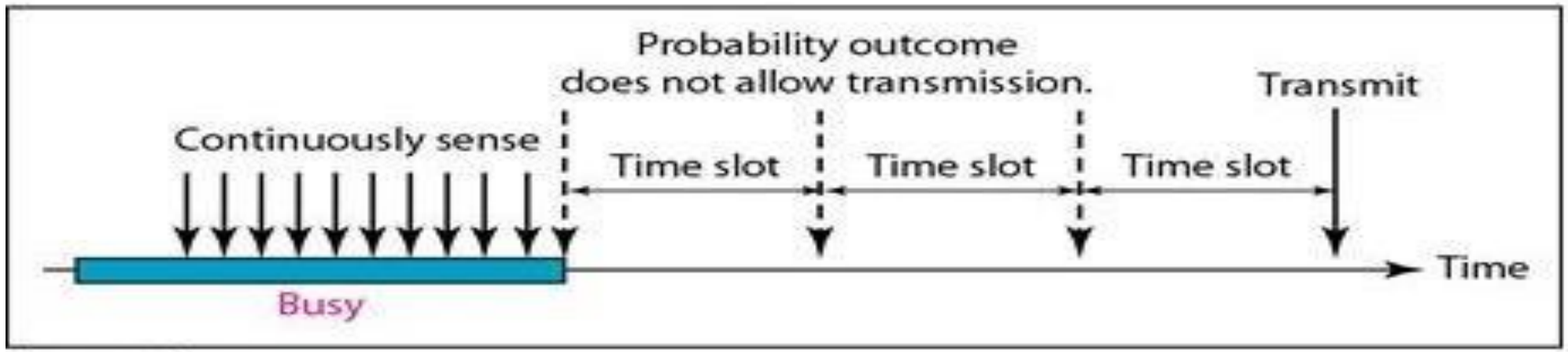
- Before sending, a station senses the channel , if No one else is sending the station begins doing so itself.
- If the channel is already in use the station **does not sense (checks) continuously** for the purpose of seizing it immediately upon **detecting the end of the previous transmission.**
- **It waits for random period of time not continuously. But in 1-persistent continuously sensing(checking)**
- In this Non –persistent CSMA when the station having a frame to transmit and find that the channel is busy , **it backs off for fixed interval of time .**
- **It checks the channel again and** if the channel is free then it transmits.
- **The back off delay is determined by the transmission time of a frame , propagation time and other systems parameters.**
- **If the channel is already in use the station does not continuously** sense it for the purpose of seizing it immediately upon **detecting the end of the previous transmission.**
- The node senses the channel, if idle it sends the data, otherwise it checks the medium after a random amount of time (not continuously) and transmits when found idle.
- **But is waits random amount period of time and again it checks for activity.**



b. Nonpersistent

P-persistent CSMA:

- **It applies to Slotted channels.**
- **When a station is ready to send it sense (checks) the channel.**
- **If it is idle , it transmits with a probability p.**
- The node senses the medium, if idle it sends the data with p probability. If the data is not transmitted ((1-p) probability) then it waits for some time and checks the medium again, now if it is found idle then it send with p probability. This repeat continues until the frame is sent.
- 1. If the channel is idle transmits the probability p ,and delay one time unit with probability (1-P).
- 2. If the channel is Busy , listen until idle repeat step 1.
- 3. If transmission is delayed one time unit repeat step 1.



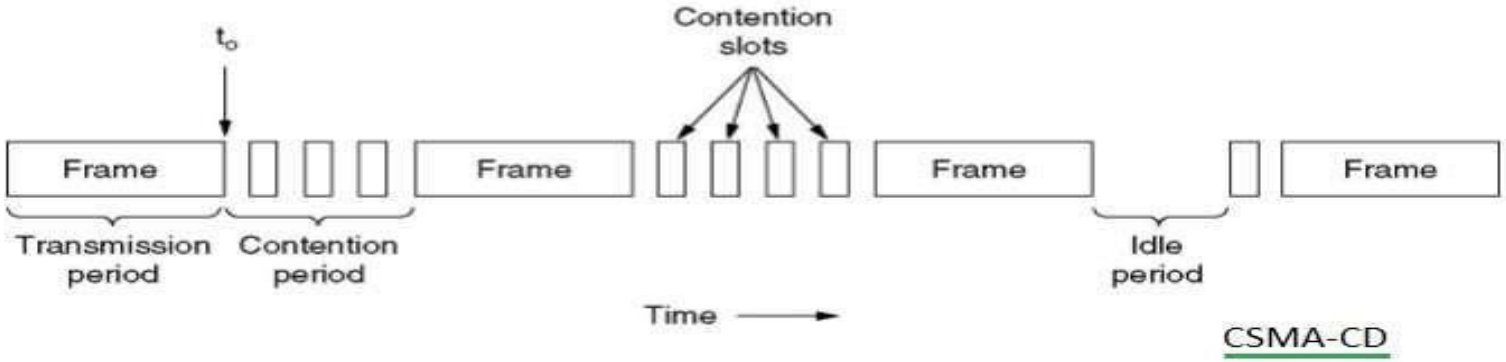
c. p-persistent

O-persistent CSMA:

- Each node is assigned at transmission order by a **supervisory node.**
- Superiority of nodes is decided before and transmission occurs in that order.
- If the medium is idle, node waits for its time slot to send data.
- All the decisions take care by the supervisory node.

(c) CSMA/CD –

- Carrier sense multiple access with collision detection.
- Stations can terminate transmission of data if collision is detected.
- Carrier Sense Multiple Access with Collision Detection (CSMA/CD) is a network protocol for carrier transmission that operates in the Medium Access Control (MAC) layer.
- It senses or listens whether the shared channel for transmission is busy or not, and defers transmissions until the channel is free.
- The collision detection technology detects collisions by sensing transmissions from other stations.
- EX: If 2 stations sense the channel to be idle and begin transmitting simultaneously , they will both detect the collision .
- Immediately stop the transmission quickly terminating the damaged frames and **Save Time and Bandwidth.**
- On detection of a collision, the station stops transmitting, sends a jam signal, and then waits for a random time interval before retransmission.



First one is **Transmission period** after immediately **Contention period** not transmission period.
Contention period Minimum time period any host to check any interrupt is occur or not.
Idle period i.e all the stations are not transmitting any data.

Fig. . At the point marked **t0 (station** finished time), a station has finished transmitting its frame. Any other station having a frame to send may now attempt to do so. If two or more stations decide to transmit simultaneously, there will be a collision. Collisions can be detected by looking at the power or pulse width of the received signal and comparing it to the transmitted signal.

After a station detects a collision, it aborts its transmission, waits a random period of time, and then tries again, assuming that no other station has started transmitting in the meantime. Therefore, our model for CSMA/CD will consist of alternating contention and transmission periods, with idle periods occurring when all stations are quiet.

The Steps for CSMA/CD is:

- When a frame is ready, the transmitting station checks whether the channel is idle or busy.
- If the channel is busy, the station waits until the channel becomes idle.
- If the channel is idle, the station starts transmitting and continually monitors the channel to detect collision.
- If a collision is detected, the station starts the **collision resolution algorithm**.
- The station resets the retransmission counters and completes frame transmission.

The Steps for Collision Resolution is:

- The station continues transmission of the current frame for a specified time along with a jam signal, to ensure that all the other stations detect collision.
- The station increments the retransmission counter.
- If the maximum number of retransmission attempts is reached, then the station aborts transmission.
- Otherwise, the station waits for a back off period which is generally a function of the number of collisions and restart main algorithm.

CSMA/CA: (Collision free protocols)

- CSMA/CA stands for Carrier Sense Multiple Access/ Collision Avoidance.
- It is a network protocol for transmission. It operates in the Medium Access Control Layer. This protocol is effective before the collision.
- CSMA/CA is effective before a collision.
- CSMA/CA is generally used in wireless networks.
- CSMA/CA minimizes the risk of collision.
- CSMA/CA initially transmits the intent to send the data, once an acknowledgment is received, the sender sends the data.
- CSMA/CA is part of the IEEE 802.11 standard.
- CSMA/CA is similar in efficiency as CSMA.

- In computer networks, when more than one station tries to transmit simultaneously via a shared channel, the transmitted data is garbled. This event is called collision.
- The Medium Access Control (MAC) layer of the OSI model is responsible for handling collision of frames.
- Collision – free protocols are devised so that collisions do not occur.
- Protocols like CSMA/CD and CSMA/CA nullifies the possibility of collisions once the transmission channel is acquired by any station.
- However, collision can still occur during the contention period if more than one stations starts to transmit at the same time.
- Collision – free protocols resolves collision in the contention period and so the possibilities of collisions are eliminated.

Types of Collision – free Protocols:

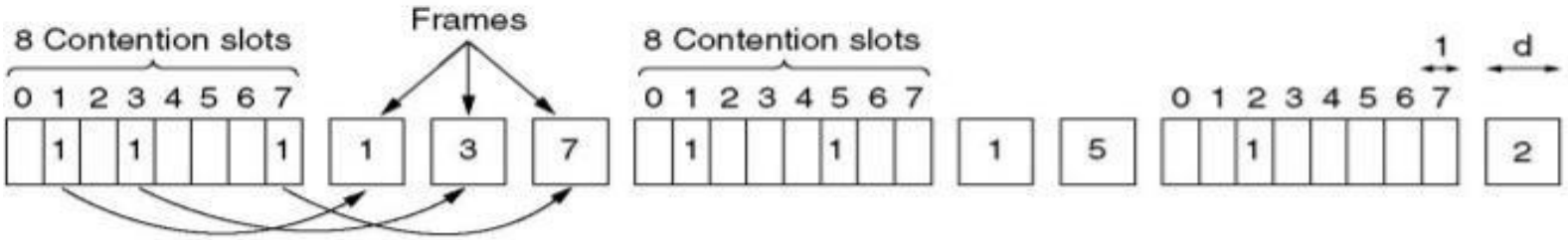
1.Bit Map Protocol.
2.Binary Count Down protocol.

Bit – map Protocol: (Reservation protocol)

- In bit map protocol, the contention period is divided into N slots, where N is the total number of stations sharing the channel.
- If a station has a frame to send, it sets the corresponding bit in the slot.
- So, before transmission, each station knows whether the other stations want to transmit.
- Collisions are avoided by mutual agreement among the contending stations on who gets the channel.
- Suppose that there are 10 stations. So the number of contention slots will be 10.
- If the stations 2, 3, 8 and 9 wish to transmit, they will set the corresponding slots to 1 as shown in the following diagram:
- Once each station announces itself, one of them gets the channel based upon any agreed criteria.
- Generally, transmission is done in the order of the slot numbers.
- Each station has complete knowledge whether every other station wants to transmit or not, before transmission starts. So, all possibilities of collisions are eliminated.

- In our first collision-free protocol, the basic bit-map method, each contention period consists of exactly N slots.
- If station 0 has a frame to send, it transmits a 1 bit during the zeroth slot.
- No other station is allowed to transmit during this slot. Regardless of what station 0 does, station 1 gets the opportunity to transmit a 1 during slot 1, but only if it has a frame queued.
- In general, station j may announce that it has a frame to send by inserting a 1 bit into slot j.
- After all N slots have passed by, each station has complete knowledge of which stations wish to transmit.
- Since everyone agrees on who goes next, there will never be any collisions.

The basic bit-map protocol



Binary Countdown:

- In Binary count down protocol all the stations on the channel are assigned a fixed length binary addresses.
- If any station wants to transmits a message or data first it broad cast its address into binary bit string . String with higher order bit.
- Different station addresses are together to decide the priority of transmitting the data.

For example: if stations 0010, 0100, 1001, and 1010 are all trying to get the channel, in the first bit time the stations transmit 0, 0, 1, and 1, respectively. These are ORed together to form a 1. Stations 0010 and 0100 see the 1 and know that a higher-numbered station is competing for the channel, so they give up for the current round. Stations 1001 and 1010 continue.

The next bit is 0, and both stations continue. The next bit is 1, so station 1001 **gives up**. The winner is station 1010 because it has the highest address. After winning the bidding, it may now transmit a frame, after which another bidding cycle starts. The protocol is illustrated in Fig. . It has the property that higher-numbered stations have a higher priority than lower numbered stations, which may be either good or bad, depending on the context.

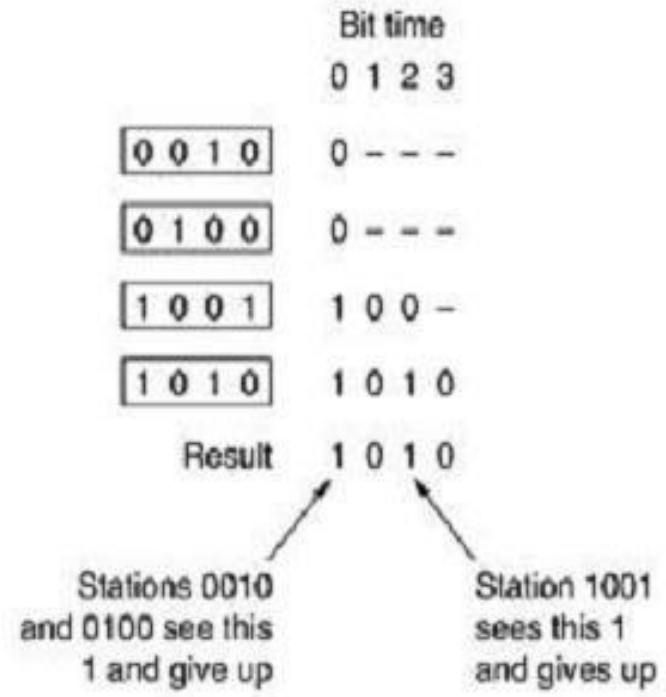
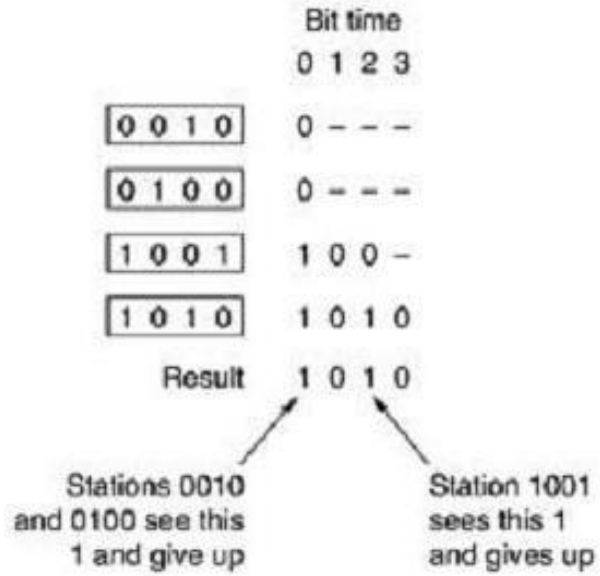


Figure . The binary countdown protocol. A dash indicates silence.



Limited Contention Protocols:

These protocols combines the advantages of collision based protocols and collision free protocols.
Under light load, they behave like ALOHA scheme. Under heavy load, they behave like bitmap protocols.

when more than one station tries to transmit simultaneously via a shared channel, the transmitted data is garbled, an event called collision. In collision based protocols like ALOHA, all stations are permitted to transmit a frame without trying to detect whether the transmission channel is idle or busy. In slotted ALOHA, the shared channel is divided into a number of discrete time intervals called slots. Any station having a frame can start transmitting at the beginning of a slot. Since, this works very good under light loads, limited contention protocols behave like slotted ALOHA under low loads.

Adaptive Tree Walk Protocol:

Adaptive Tree Walk Protocol is a technique for transmitting data over shared channels that combines the advantages of collision based protocols and collision free protocols.

In computer networks, when more than one station tries to transmit simultaneously via a shared channel, the transmitted data is garbled, an event called collision. In collision based protocols like ALOHA, all stations are permitted to transmit a frame without trying to detect whether the transmission channel is idle or busy. This works very good under light loads. Under heavy loads, collision free protocols are suitable, since channel access is resolved in the contention period that eliminates the possibilities of collisions.

In adaptive tree walk protocol, the stationed are partitioned into groups in a hierarchical manner. The contention period is divided into discrete time slots, and for each slot the contention rights of the stations are limited. Under light loads, all the stations can participate for contention each slot like ALOHA. However, under heavy loads, only a group can try for a given slot.

Sr. No.	Key	CSMA/CA	CSMA/CD
1	Effectiveness	CSMA/CA is effective before a collision.	CSMA/CD is effective after a collision.
2	Network Type	CSMA/CA is generally used in wireless networks.	CSMA/CD is generally used in wired networks.
3	Recovery Time	CSMA/CA minimizes the risk of collision.	CSMA/CD reduces recovery time.
4	Conflict Management	CSMA/CA initially transmits the intent to send the data, once an acknowledgment is received, the sender sends the data.	CSMA/CD resends the data frame in case a conflict occurs during transmission.
5	IEEE Standards	CSMA/CA is part of the IEEE 802.11 standard.	CSMA/CD is part of the IEEE 802.3 standard.
6	Efficiency	CSMA/CA is similar in efficiency as CSMA.	CSMA/CD is more efficient than CSMA.